Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation: a staged simulation study and a recommended analysis for the Hybrid design

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to obtain a correctly sized test, and whether to guard against a prognostic biomarker.

Methods. We ran an exploratory, staged simulation study under the data-generating process of a companion paper. There the interaction is a biomarker-drug-response correlation. Stage 1 compares a binary on-drug indicator (G1), that indicator plus a lagged-treatment term (G2), and an exposure-weighted predictor at the true half-life (G3) over 162 configurations. Stage 2 selects the half-life by AIC (G4). Stage 3 adds CR2 cluster-robust inference (G8, G9) and measures the cost of selection. Stage 4 adds the original analysis (E0) and an expectancy-term family (F1, F1A0B). Stage 5 stresses G9 with three prognostic correlation pathways, dropout and random slopes.

Results. The lagged term changes nothing. Exposure weighting recovers most of the power that carryover removes under the Hybrid and OL+BDC designs. AIC selection matches the power at the true half-life without recovering it. The model-based test rejects at about 0.037. CR2 brings every specification to within half a point of nominal. Selection adds about half a point. Pooled over 7,987 null fits, G9 rejects at 0.055 against 0.049 for G8 (paired difference 0.006, SE 0.001). The G9 interval covers at about 0.93 to 0.94 under the alternative. E0 is liberal under the crossover and OL+BDC designs and marginally liberal under Hybrid. G9 holds size under the three prognostic pathways at every feasible strength and loses no power to F1A0B. A power loss at the strongest confound came from an infeasible covariance matrix that the code had repaired. A closed-form feasibility bound now excludes such scenarios.

Conclusions. For a Hybrid trial under this data-generating process we recommend G9: the exposure-weighted predictor with an AIC-selected half-life, refitted by REML and tested with CR2. Its size is about 0.055 at $N \leq 70$ and its interval is slightly narrow. A fixed compromise half-life would avoid that cost and is under evaluation. The recommendation is conditional on the process and parameter set simulated.

Keywords: N-of-1 trials; carryover; predictive biomarker; cluster-robust inference; post-selection inference; simulation study.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator. The within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series. Pooling allows population-level inference at sample sizes smaller than a parallel-group trial would require (1). The estimand here is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We evaluate analysis approaches for this estimand in the Hybrid N-of-1 design defined in Section 2.2.

The main inferential complication in N-of-1 trials is carryover. A drug's pharmacological effect does not stop at discontinuation. It decays as a function of the drug's half-life. Observations taken after discontinuation therefore still carry part of the treatment effect. An exposure regressor that ignores this decay is a mismeasured version of what the patient received. The consequences are twofold. First, the interaction contrast is attenuated. Occasions that still carry a large fraction of the drug effect are coded the same as occasions that carry none. This biases the biomarker-by-exposure coefficient toward zero, as any covariate measured with error would. Second, those occasions vary in a way the model does not represent. The unexplained variation inflates the standard error of the same coefficient. Power falls for both reasons.

This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, there is no biomarker-response association at any level of residual exposure. A miscoded regressor then has no interaction to distort, and test size does not depend on the exposure coding. The conservatism reported in Section 3.3 has a separate origin in the working covariance and applies equally to every specification. Carryover misspecification is therefore a question of efficiency. The first stage of the study is framed in terms of recovered power.

A companion study (2) builds on the data-generating process of Hendrickson et al. (1). In that process the interaction is encoded as a correlation between the biomarker and the drug response. The correlation is present on drug and decays off drug: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$. We call this the correlation encoding of the interaction. The companion study shows that carryover reduces power under this encoding, and that the reduction depends on the design. The loss under a given analysis is governed by the on/off correlation difference, that is, the on-drug correlation minus the mean off-drug correlation, which the analysis's exposure coding determines. At a half-life of one week in this study's Stage 1 grid, the binary indicator loses a fifth to two thirds of its power under the Hybrid and OL+BDC designs. It loses a tenth to a sixth under the crossover design (Section 3.1). We restrict attention to the correlation encoding. The carryover problem is largest there, and the original results of Hendrickson et al. were obtained there. That earlier work fixed the decay shape at exponential and evaluated only one analysis specification. Simulation studies of aggregated N-of-1 analyses under carryover exist (3–6). One includes an exponentially decaying wash-in and wash-out term (6). All target the treatment effect rather than the biomarker-by-treatment interaction, which is the estimand here. An interaction test needs about four times the sample of a main-effect test of the same size (7). The correlation encoding is a modeling assumption of this project, not a validated description of any trial. Every recommendation below is conditional

on it.

This manuscript reports the sequence of decisions by which we arrived at a recommended analysis for the Hybrid design. It is not a single cross-sectional comparison. The analysis model was built in stages. Each stage answered a question the previous one raised. The order of the stages is the order in which the questions arise for an analyst.

The first question is how residual exposure should enter the linear predictor. Three positions are available toward the unknown decay. The analyst may ignore it and code exposure as on or off (Unadjusted, G1). G1 is the benchmark any remedy must beat. The analyst may estimate it by adding a lagged just-off-drug nuisance term whose magnitude the data determine (Lag-adjusted, G2). G2 is the classical crossover-trial device of Jones and Kenward (8). Or the analyst may assume it by committing to a decay form and half-life in a continuous exposure-weighted predictor (Exposure-weighted, G3). G1 and G2 estimate the same coefficient and differ by exactly one nuisance term, so their contrast isolates that term. G3 changes the regressor that carries the interaction. We cross all three against exponential and Weibull data-generating decay forms in a full factorial (Stage 1).

The second question follows from the third position. An assumed half-life is the true half-life, which no real analyst knows. We therefore let the data choose it by AIC over a candidate grid (G4) (Stage 2).

The third question concerns the size of the test rather than its power. The working covariance of the mixed model is a random intercept with an AR(1) residual correlation. It does not match the covariance that the data-generating process induces, and the model-based test is conservative. A CR2 bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom (9) corrects this. We apply the correction to each specification and check separately that data-driven half-life selection does not inflate size (Stage 3).

The fourth question is whether anything outside this family does better. Two comparators are fitted to the same simulated trials. The first is the original analysis (E0), which uses a binary indicator and no residual correlation structure. The second is a richer family developed in parallel within this program (F1, F1A0B). That family adds design-expectancy terms to protect against a prognostic biomarker. It either fixes the half-life or calibrates its selection by a parametric bootstrap (Stage 4).

The fifth question is the robustness of the leading specification to the conditions a real trial is most likely to violate. These are a biomarker that is prognostic as well as predictive, through three correlation pathways at two strengths, and a random-effects structure richer than a random intercept (Stage 5). This stage also produced a methodological finding about the simulation itself. A non-positive-definite covariance matrix had been repaired without warning. We report it because it changes how such stress tests must be designed.

Two defects in the simulation code were found during the study and corrected. The first was a pre-exposure coding error in the exposure-weighted predictor that affected the crossover design. The second was a cross-factor correlation constant that departed from the documented data-generating process. Every result in this manuscript was produced under the corrected code, with one labeled exception. We follow the ADEMP reporting framework of Morris,

White, and Crowther (10) throughout. We also follow later reporting guidance on common random numbers and on comparative simulation studies (11,12). The study is exploratory. Its stages were not fixed in advance, and the recommendation is the end of a documented sequence of choices rather than a pre-specified comparison. Section 2 describes the simulation design in ADEMP order. Section 3 gives the results stage by stage. Section 4 states the recommendation and its limits.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with three latent response components. These are the drug-response component, the natural-history component, and the placebo-response component. The four quantities are drawn jointly across all eight measurement occasions from a multivariate normal distribution. The means follow modified Gompertz trajectories. The covariance matrix contains within-component AR(1) autocorrelation, cross-component correlations, and the biomarker-response correlation that encodes the interaction and depends on treatment state (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity minus the sum of the three components. A beneficial effect therefore lowers the score. Carryover enters through a decay function of time since discontinuation. The decay keeps part of the drug-response component present at nominally off-drug occasions. Every analysis specification in a given stage is then fitted to the same simulated dataset. We record the interaction coefficient, its standard error, and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.6. It is repeated 1,000 times in the random-effects comparison of Stage 5 and in the large-sample cells of the supplementary study, and 2,000 times in the supplementary size diagnostic.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired. Paired differences have substantially less Monte Carlo variance than differences between independent draws. Throughout, N denotes the **total** number of patients in a trial. Patients are allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim is to arrive at a recommended analysis specification for the biomarker-treatment interaction in a Hybrid aggregated N-of-1 trial under the correlation encoding. A second aim is to document the evidence at each stage. The stages address five questions in order. The first is how residual exposure should enter the linear predictor. The second is whether the analyst's assumed half-life can be replaced by a data-driven choice. The third is whether the test is correctly sized and how to make it so. The fourth is whether the original analysis or a

richer expectancy-term family does better on the same trials. The fifth is whether the leading specification holds up under a prognostic biomarker and a richer random-effects structure.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged. They therefore differ in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient. Each randomization path is a fixed on-drug/off-drug schedule. Each design also carries an expectancy weight, defined as the participant's probability of being on drug by design. The weight is one during open-label periods and one half during blinded periods, and it scales the placebo-response component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout. Occasions are spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions and then on comparator. The other path reverses that order, so it never discontinues and contributes no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The design of Hendrickson et al. and this program's reference design. It consists of open-label titration at weeks 4 and 8, blinded discontinuation weekly from week 9 to 12, and a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It has the most on/off transitions of the three designs. It is the only one that allows more than one discontinuation per patient. The recommendation of this manuscript is made for this design. CO and OL+BDC are carried through Stages 1 to 4 to show where the recommendation does and does not transfer.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks after switching under CO. At the half-lives examined here, roughly half the drug effect persists into the first off-drug occasion under the two non-crossover designs. Less than a fifth persists under CO. The designs therefore differ substantially in how much carryover they can accumulate.

2.3 Data-generating mechanism

Under the correlation encoding, the biomarker and the drug-response component are drawn jointly from a multivariate normal distribution. Their correlation depends on treatment state: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} is the time since drug discontinuation. The data-generating process is the extended DGP of the companion study. The extended DGP differs from the original DGP of Hendrickson et al. in four respects. The biomarker-response correlation is zero before first exposure and decays after discontinuation at the carryover half-life. The within-component correlation is a separable AR(1) structure at $\rho = 0.7$. That is, every component has the same first-order autoregressive correlation, and

a single cross-component constant $c_1 = 0.2$ is scaled by $\rho^{|w_t - w_s|}$ at every lag. Carryover on the drug-response mean decays from the last on-drug mean. A covariance matrix that is not positive definite stops the simulation rather than being repaired (Section 2.8).

The decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. Two forms are used. Each is matched to a common half-life $t_{1/2}$. Exponential decay,

$$\phi_{\text{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}},$$

is classical first-order elimination and the default pharmacokinetic assumption. Weibull decay,

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}},$$

has a shape parameter k. The elimination tail is heavier than exponential when $k < 1$ and lighter when $k > 1$. The form reduces to exponential at $k = 1$. The Stage 1 factorial uses $k \in \{0.7, 1.5\}$ alongside exponential. The Stage 4 decay-shape misspecification cells use $k \in \{0.5, 2\}$. All forms pass through the same half-life-defining point, $\phi = 0.5$ at $t_{sd} = t_{1/2}$. They are therefore comparable at a fixed nominal half-life even though their shapes differ on either side of it. The decay form the analysis assumes differs by stage. In Stage 1 the exposure-weighted predictor is built with the data-generating form, shape and half-life. This is the driver's default when no analysis form is given. The Weibull cells of Stage 1 therefore test the representation under a correctly specified decay, not its robustness to the shape. Every analysis from Stage 2 onward assumes exponential decay. The Weibull cells of Stage 4 are the shape-misspecification check.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale. That scale is the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$. Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The base analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))`. The specifications differ in how X is constructed, whether a nuisance term is added, and how the standard error is obtained. In the code and data files described in the Reproducibility section, each specification is stored under an `E`-prefixed code. This manuscript refers to them by the `G1-G9` code of Table 1, or by name. Six of the nine carry the argument: G1, G2 and G3 (Stage 1), G4 (Stage 2), and G8 and G9 (Stages 3 to 5). G5, G6 and G7 appear in the nine-specification heatmaps of Stage 3 for completeness and are otherwise not discussed.

Table 1: The nine G-family analysis specifications. G1-G3 are the Stage 1 representations of residual exposure; G4 replaces the assumed half-life by a data-driven choice (Stage 2); G6-G9 are CR2 cluster-robust variants of G1, G2, G4 and G3 (Stage 3). G5 collapses to one row per patient before fitting and has no CR2 variant. The external comparators E0 and the F-family are defined in Section
refsec-comparators.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

2.4.1 Unadjusted (G1): binary on-drug indicator

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented. An off-drug occasion that still carries most of the drug effect is coded the same as one that carries none. This is the default an analyst obtains without considering carryover. We treat it as the benchmark against which the other specifications must justify their additional structure. It uses no half-life, so it does not depend on the analyst's pharmacokinetic assumptions.

2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

We retain $X = D_{it}$ and add a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$. It marks off-drug timepoints that immediately follow an on-drug one. This is the classical crossover-trial device of Jones and Kenward (8). Carryover magnitude is estimated from the data rather than assumed. Like Unadjusted, this specification does not depend on the assumed half-life. L_{it} enters only as a nuisance main effect. There is no bm:L_{it} product, so the interaction tested remains bm:D_{it}. Lag-adjusted can absorb mean-level carryover. It cannot correct the attenuation of the interaction.

2.4.3 Exposure-weighted (G3): continuous decayed predictor

Here $X = D_{bc,it}$, a continuous exposure-decayed predictor. It is built from the analyst's assumed decay form and analysis half-life $t^a_{1/2}$. At on-drug timepoints $D_{bc,it} = 1$. Off drug after first exposure it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/t^a_{1/2}}$. It loses half its value every $t^a_{1/2}$ weeks after discontinuation. Before any exposure $D_{bc,it} = 0$. This pre-exposure rule was absent in earlier drafts (Section 2.8). The half-life assumption enters only through the predictor values passed to `lme`. Exposure-weighted is therefore a different model for each assumed half-life. As $t^a_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is the boundary member of the Exposure-weighted

family at zero assumed half-life. In Stage 1 the assumed half-life is set equal to the data-generating one. This is the true half-life, which no real analyst knows. Stage 2 removes that assumption. Under the correlation encoding, the regression of the response on the biomarker at an occasion is linear in $c_{bm}\,\phi(t_{sd})$. $D_{bc,it}$ at the true ϕ is therefore the correctly specified regressor. Stage 1 compares a correct model with two incorrect ones. The question Stage 1 answers is how much the incorrect ones lose.

2.4.4 AIC-selected half-life (G4)

This specification uses the Exposure-weighted formula but chooses the analysis half-life $\widehat{t}^{\,a}_{1/2}$ from the data. Write $M(t^a_{1/2})$ for the `lme` fit of the full formula, interaction included, with $D_{bc,it}$ constructed at a candidate half-life. The selected half-life is

$$\widehat{t}^{\,a}_{1/2} = \underset{h\,\in\,\mathcal{T}}{\arg\min}\,\mathrm{AIC}(M(h)), \qquad \mathcal{T} = \{0.25,\ 0.5,\ 1.0,\ 2.0\}\ \text{weeks},$$

with $\mathrm{AIC}(M) = -2\log L(M) + 2p$ (13). The candidate fits are compared by maximum likelihood. The candidates differ in a fixed-effects column, and REML likelihoods are not comparable across fixed-effects designs. The selected model is then refitted by REML. The interaction coefficient, its standard error and its p-value are read from that single refit, not pooled across $\mathcal{T}$. This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own real-data half-life-selection routine. The selection model contains the interaction term, so the selected half-life depends on the coefficient under test. In the vocabulary of the F-family below this is the F1A construction, not F1A0. The cost is a post-selection inference risk (14–16). The reported standard error treats $\widehat{t}^{\,a}_{1/2}$ as fixed in advance when the same data chose it. A test that is correctly sized under one configuration need not be correctly sized under all. Stage 3 measures what the selection costs in size. The paired reference is the same predictor at the true half-life (G8). Stage 3 also reports the distribution of the selected half-life.

2.4.5 CR2 cluster-robust variants (G8, G9)

The standard error of $\beta_{bm:X}$ read from the fitted `lme` object is valid only if the assumed residual covariance is correct. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation. The companion calibration study (17) shows that this does not match the covariance the data-generating process induces. We quantify the consequence with the calibration ratio

$$\kappa = \frac{\overline{\widehat{SE}}}{SD(\widehat{\beta}_{bm:X})},$$

the mean model-reported standard error across replicates divided by the empirical standard deviation of the replicate point estimates. $\kappa = 1$ is correctly calibrated and $\kappa > 1$ is a

conservative test. The mean of the standard errors is below their root mean square. κ as defined is therefore biased downward by a few percent when the standard errors vary across replicates, as they do at $N = 35$. The empirical SD in the denominator is also sensitive to a heavy-tailed estimate distribution. Section 3.4 meets that case under heavy missingness and reports a robust ratio beside it. Coverage of the nominal 95% interval is the complementary measure. It is reported where the stored per-replicate output allows it. The remedy for the mismatch is a cluster-robust sandwich variance. It derives the variance of $\hat{\beta}_{bm:X}$ from observed between-patient variation rather than from the assumed covariance. With only $N = 35$ or 70 clusters the naive Liang-Zeger form is itself biased downward. We therefore use the CR2 bias-reduced estimator of Bell and McCaffrey (9) with Satterthwaite degrees of freedom, as implemented in `clubSandwich` (18). It replaces only the standard error, the reference distribution and the p-value. Point estimates are unchanged. The Kenward-Roger correction (19) is the standard model-based small-sample alternative. It corrects the degrees of freedom and the variance under the working covariance, so it does not address a misspecified one. It is not evaluated here. G8 is G3 with CR2 inference and G9 is G4 with CR2 inference. G6 and G7 are the corresponding variants of G1 and G2. G6 and G8 share G9's inference, so they are the like-for-like comparators for its power. The model-based G1 and G3 are conservative. Part of any power difference against them is the increase in power from correcting a conservative test.

2.4.6 External comparators: the original analysis (E0) and the F-family

Two further analyses are fitted alongside the G-family to the same simulated trials in Stages 3 to 5.

E0 is the original analysis published with the original DGP (1). We read it from the original repository (commit `58b32a9`, `R/lme_analysis.R`) rather than from this project's vendored copy, which had acquired a residual correlation structure at some later point. It fits `lme4::lmer` with a random intercept and no residual correlation structure. It uses the binary on-drug indicator and takes Satterthwaite p-values from `lmerTest`. As published it also adds the expectancy main effect D_e whenever expectancy varies within the design, and it drops patients who are never on drug. E0 therefore differs from G1 in three respects: no `corCAR1`, the D_e term, and the software and degrees-of-freedom method. The comparison does not isolate any one of them.

The F-family adds the design-expectancy covariate D_e and its product with the biomarker to the G3 formula. D_e is η_t, one during open-label periods and one half during blinded periods. The formula is $\text{Sx} \sim bm + t + D_{bc} + bm{:}D_{bc} + D_e + bm{:}D_e$, with the same random intercept, `corCAR1` and CR2 inference as G8 and G9. The expectancy terms were introduced in the companion study to protect the interaction test against a biomarker that is prognostic rather than predictive. The members differ only in how the analysis half-life is set. **F1** fixes it at the project's policy value of 0.2 week. That value is below every data-generating half-life used in Stages 2 to 5 (0.5, 1 and 2 weeks). Every G9-against-F1 contrast at those cells is therefore in part a contrast between a selected and a misspecified half-life. **F1A** selects the half-life by AIC with the interaction in the selection model, as G4 does. **F1A0** selects it

on the model without the interaction and refits with it. **F1A0B** calibrates the F1A0 test by a parametric bootstrap under the null that repeats the selection on every resample. The F1A0B candidate grid is $\{0, 0.2, 0.5, 1, 2\}$ weeks, wider than that of G4. Its bootstrap is a single-resample bootstrap: one null resample per Monte Carlo replicate, with the critical value estimated from the pooled resamples of the cell. Its calibration therefore carries the Monte Carlo error of the cell. F1 and F1A0B are the two reported here. D_e is constant within every path of the crossover design. The F-family is therefore not identifiable there and is reported for Hybrid and OL+BDC only.

2.4.7 Prognostic biomarker pathways

A biomarker may be associated with the outcome through channels other than the drug response. An interaction test that rejects because of such an association is not usable in practice. Three pathways are implemented as entries in the biomarker row of the joint correlation matrix. Each sets one parameter with all others at their defaults. The first is a correlation between the biomarker and baseline severity (prog-BL, c_{bl}). The second is a correlation with the natural-history component at every occasion (prog-TV-const, c_{tv}). The third is a correlation with the placebo-response component at every occasion (prog-PB, c_{pb}). We refer to these as prognostic correlations and use the short labels in tables and prose from here on. Their feasible strengths are bounded, and Stage 5 derives the bound. A fourth setting is an additive shift of the natural-history mean proportional to the standardized biomarker (b_{tv}, labeled `prog-TV-grow` in the stored data). Despite its label, the shift is constant over occasions. It is absorbed exactly by the biomarker main effect of every analysis, so it cannot move the interaction. It serves only as a check of that absorption. A biomarker that predicts the rate of natural-history change produces a shift that grows with time and is not bounded by the correlation matrix. That pathway is the one most likely to bias an interaction estimator whose model has no biomarker-by-time term. It is not among the simulated pathways. The Discussion returns to it.

2.5 Choice of analysis specifications

The specifications were chosen to represent distinct, established positions an analyst can take toward an unmeasured decay, and then to answer the specific questions each stage raised. They were not chosen to survey the wider space of N-of-1 analysis models. Bayesian hierarchical models, GEE as a primary model, randomization tests, flexible spline decay curves, state-space treatments of drug concentration and two-stage meta-analytic combination are all established in the N-of-1 or crossover literature. None is evaluated here. Section 4.5 returns to three of them. An earlier draft also tested carryover terms interacted with the biomarker (`bm:L` and `bm:t_{sd}`). Both had lower power than G1 at every cell. We attribute this to the data-generating process having a single true decay parameter, so that a second interacted coefficient adds estimation noise with no signal to recover. They were dropped in favor of G4, which addresses the same unknown, the decay half-life, by estimating it.

2.6 Factorial grid and parameter settings

Stage 1 crosses biomarker moderation, carryover half-life, trial design, sample size and decay form against G1, G2 and G3:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, Hybrid, OL+BDC
Decay form	exponential; Weibull $k \in \{0.7, 1.5\}$
Replicates per cell	500

Three half-lives, three designs, two sample sizes, three biomarker levels and three decay forms give 162 distinct configurations in the correlation-encoding slice reported here. The production grid holds 540 cells per encoding. These are the 162 configurations, a Weibull $k = 1$ condition (54 cells), a linear decay form (216 cells), and a fourth specification (stored as `E4`) from an earlier draft. The Weibull $k = 1$ condition is the exponential form drawn with its own random numbers, so it is a replicate-level duplicate rather than a numerical one. The linear decay form is not pharmacokinetically motivated and is not reported. The `E4` specification is not part of this study. Counts in the text refer to the 162 distinct configurations unless the 216 exponential-plus-Weibull cells are named. The $c_{bm} = 0$ row is the Type I error check against nominal 5%. The $t_{1/2} = 0$ row gives a best-case power reference at each combination of N, design and c_{bm}.

Stages 2 to 4 use a 30-cell grid on which every specification is fit to the same trial each replicate. The grid uses exponential decay, design $\times$ {CO, Hybrid, OL+BDC}, and $N \in \{35, 70\}$. Panel A varies $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1$ week. Panel B varies $t_{1/2} \in \{0, 0.5\}$ at $c_{bm} = 0.45$. Stage 4 adds thirteen scenario cells on Hybrid at $N = 70$ and $t_{1/2} = 0.5$ week. These are the null and two alternatives ($c_{bm} = 0.45$, 0.25); the OL+BDC null and alternative; the three prognostic correlation pathways and the constant-shift setting at strength 0.3 under the null; a four-fold half-life misspecification (true $t_{1/2} = 2$ weeks, null and $c_{bm} = 0.25$); and a decay-shape misspecification (Weibull $k = 2$ data, exponential analysis, null and $c_{bm} = 0.25$). Stage 5 crosses the same four prognostic settings with strength $\{0.3, 0.6\}$, $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$ on Hybrid at $t_{1/2} = 0.5$ week (32 cells, 500 replicates). It also refits G9 with three random-effects structures on twelve of those cells at 1,000 replicates.

A supplementary study, specified in a working protocol after Stage 5 (`32-gap-fill-protocol.md`), then closed the items the stages left open. All of its cells are on Hybrid with every specification fit to the same trial, at 500 replicates unless stated. It replaces the size diagnostic with four corrected-code null cells at 2,000 replicates ($t_{1/2} \in \{0.5, 1\}$ weeks, $N \in \{35, 70\}$). It extends two Stage 4 scenarios from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. The first is Weibull $k \in \{0.5, 2\}$ data analyzed as exponential ($c_{bm} \in \{0, 0.25, 0.45\}$). The second is a true half-life of 0.2, 2 or 4 weeks ($c_{bm} \in \{0, 0.25\}$).

The value 0.2 is below the G4 candidate grid, 4 is above it, and 2 is at its boundary. The study adds a third scenario, the within-component autocorrelation $\rho \in \{0.5, 0.9\}$, at $c_{bm} \in \{0, 0.30\}$. The largest feasible c_{bm} falls with ρ, and 0.45 is infeasible at $\rho = 0.9$ (feasibility bound 0.391). The study also runs a prognostic ladder at $N = 70$ chosen inside the feasibility bounds of Section 3.5. The ladder sets c_{pb} and c_{tv} at 0.40 and 0.45 and c_{bl} at 0.6 and 0.8, each at $c_{bm} \in \{0, 0.30\}$, plus $c_{bl} = 0.45$ at $c_{bm} \in \{0, 0.45\}$. No prog-PB or prog-TV-const cell is run at $c_{bm} = 0.45$ above the Stage 5 strength of 0.3, since the feasibility bound there is 0.306. The ladder also runs prog-PB at $c_{pb} \in \{0.3, 0.45\}$ with $c_{bm} = 0$ at $N \in \{140, 280\}$ (1,000 replicates). Finally, the study imposes monotone dropout at rates of 0.1, 0.2 and 0.3, with $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$. Dropout is either completely at random (MCAR) or has a hazard that rises with the baseline severity rank (MAR). The MAR hazard is scaled by twice the rank so that the average hazard is unchanged. The rate is the fraction of patients whose series is incomplete. The first occasion is always observed. The per-occasion hazard over the remaining seven is $1 - (1 - \text{rate})^{1/7}$ (0.015, 0.031 and 0.050), which keeps about 95, 90 and 84 percent of the observations. A first run used the earlier convention of the driver, in which the rate is the hazard at each of the eight occasions including the first. That convention keeps 64, 42 and 27 percent of the observations and leaves 10, 20 and 30 percent of patients with none. It is therefore not a 10 to 30 percent dropout scenario. It is reported separately as a heavy-missingness stress case. Every cell passed a pre-flight check that the smallest eigenvalue of its covariance matrix is positive before any simulation ran.

2.7 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al. (10, Tables 5-6). We chose $n_{sim} = 500$ so that the MCSE on power at $p = 0.5$ would not exceed 0.022. At $n_{sim} = 500$ the two-MCSE band around a nominal 0.05 rejection rate is approximately $[0.031, 0.069]$. That band is a check on gross inflation, not a demonstration of nominal size. A single 500-replicate cell whose true size is 0.065 falls inside the band with probability about 0.64. One whose true size is 0.075 falls inside it with probability about 0.31 (normal approximation). Claims about size therefore rest on the pooled 2,000-replicate cells of Stage 3 and on paired contrasts between specifications. Single-cell band membership is reported as such. No simulation runs while this manuscript renders. All specifications are fit to the same simulated dataset within each cell (common random numbers). Every pairwise contrast between specifications therefore has reduced Monte Carlo variance relative to independent draws. Where a paired standard error is quoted, it is computed from the per-replicate rejection indicators. The drivers also seed replicate r of every cell from a job-level base plus r. Cells at the same design and N within a job therefore share a random-number stream. Contrasts between such cells are paired too. Counts of cells inside a band are not independent confirmations and are not presented as such. Comparisons across jobs, and across the two simulation hosts, are unpaired and carry the full Monte Carlo error of both cells. The two hosts do not produce identical draws. Exact file paths and versions are given in the

Reproducibility section.

2.8 Corrections to the simulation code during the study

Two defects in the simulation driver were found and corrected while this study was in progress. A third change in policy followed from Stage 5. Each is stated here so that the reader can tell which results it affects.

1. **Pre-exposure coding of the exposure-weighted predictor (corrected 2026-10-06).** Before any exposure, $D_{bc,it}$ was coded as $\phi(0) = 1$ rather than 0. A patient on the placebo-first path of the crossover design was therefore treated as fully exposed at every visit. The same correction replaced the REML AIC comparison of G4 with the ML comparison and REML refit of Section 2.4. A paired check fitted each simulated trial under the old and new coding (`26-co-dbc-fix-paired.R`). It showed that the entire collapse of the exposure-weighted family under the crossover design reported in earlier drafts was this defect. With the correction, G3 and G4 match G1 under CO. Hybrid cells were unaffected, because no patient starts on placebo there.
2. **Cross-factor correlation constant (corrected 2026-10-21).** The documented data-generating process uses a single cross-component correlation $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag. The driver had used 0.1 at lags greater than zero. The Stage 1 factorial was rerun in full under the corrected constant. Every Stage 2 to 5 comparison was run after it. No result from before this correction is reported. The 3,000-replicate size diagnostic of earlier drafts was replaced by the 2,000-replicate corrected-code diagnostic of Section 3.3, which reproduces its conclusion.
3. **Non-positive-definite covariance matrices are refused, not repaired (policy change, 2026-10-26).** The original DGP repairs a non-positive-definite covariance matrix with `corpcor::make.positive.definite` and continues. The extended DGP checks feasibility and stops. Stage 5 found that a prognostic confound at strength 0.6 produced such a matrix and that the repair had removed part of the interaction signal. The driver now refuses such cells by default. Section 3.5 gives the evidence and the closed-form feasibility bound that replaces trial and error.

The exploratory sensitivity blocks of earlier drafts predate both corrections and are no longer part of this manuscript. Those blocks covered within-component autocorrelation, analyst-truth half-life mismatch, dropout, the effect-size curve, and single-cell comparisons of G4, G5 and the CR2 variants. Their material remains in the supplement and the archived drafts. The autocorrelation, half-life and dropout scenarios were rerun under the corrected code in the supplementary grids of Section 3.4. The dropout block of those drafts used the per-occasion convention described in Section 2.6. Under that convention a stated rate of 0.1 to 0.3 removed 36 to 73 percent of the observations. Its rates are therefore not comparable with the patient-level rates reported here.

Table 2: Stage 1: performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is mean(estimate $-\ \theta_{\text{true}}$) on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$; see the estimand-mismatch caveat in the text for G1/G2. Empirical SE is the standard deviation of the point estimates across replicates. Coverage is the proportion of nominal 95% Wald CIs containing θ_{true}. MSE is bias^2 + Empirical SE^2. Non-conv is the fraction of replicates where 'nlme::lme()' failed to converge. MCSEs follow Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.0	Lag-adjusted	0.994 (0.003)	-0.022 (0.031)	0.698	0.990 (0.004)	0.487 (0.029)	0.000
0.0	Exposure-weighted	0.994 (0.003)	-0.021 (0.031)	0.699	0.990 (0.004)	0.489 (0.029)	0.000
0.5	Unadjusted	0.946 (0.010)	+0.618 (0.034)	0.760	0.938 (0.011)	0.959 (0.055)	0.000
0.5	Lag-adjusted	0.944 (0.010)	+0.619 (0.034)	0.762	0.936 (0.011)	0.964 (0.055)	0.000
0.5	Exposure-weighted	0.962 (0.009)	+0.069 (0.039)	0.873	0.970 (0.008)	0.766 (0.052)	0.000
1.0	Unadjusted	0.806 (0.018)	+1.143 (0.036)	0.799	0.790 (0.018)	1.943 (0.092)	0.000
1.0	Lag-adjusted	0.808 (0.018)	+1.143 (0.036)	0.808	0.786 (0.018)	1.959 (0.092)	0.000
1.0	Exposure-weighted	0.888 (0.014)	-0.084 (0.047)	1.062	0.964 (0.008)	1.135 (0.074)	0.000

3 Results

Results are reported in the order the model was built. Each stage states the question it answers, the data it uses, and what it passes to the next stage. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$. Differences smaller than this should not be over-interpreted.

3.1 Stage 1: representing residual exposure

Stage 1 asks which of the three representations of residual exposure recovers the power that carryover removes. Table 2 reports performance at the reference cell (exponential DGP, Hybrid, $N = 70$, $c_{bm} = 0.45$). Non-convergence was zero there. Bias, MSE and coverage are computed against the calibrated target of Exposure-weighted, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$. Unadjusted and Lag-adjusted estimate a different coefficient, bm:D_{it}. Their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators. Those columns are comparable within the G1/G2 pair but not across to G3. For the three-way comparison we rely on power and Type I error, which are properties of the test rather than of the coefficient tested.

Stage 1 produced three findings. Figure 1 shows all three.

The lagged-treatment term changes nothing. The Unadjusted and Lag-adjusted curves coincide in every panel and at every half-life. Across the 162 distinct configurations of the correlation-encoding slice, the mean power difference is +0.002. The largest absolute difference is 0.022, against a single-cell MCSE of about 0.02. Bias, MSE and coverage agree on average. Their cell-level differences are at most 0.018, 0.081 and 0.026, all within Monte Carlo noise. The two specifications share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term. The comparison therefore isolates that term. The mechanism follows from where the term enters. It enters only as a main effect, so the tested interaction

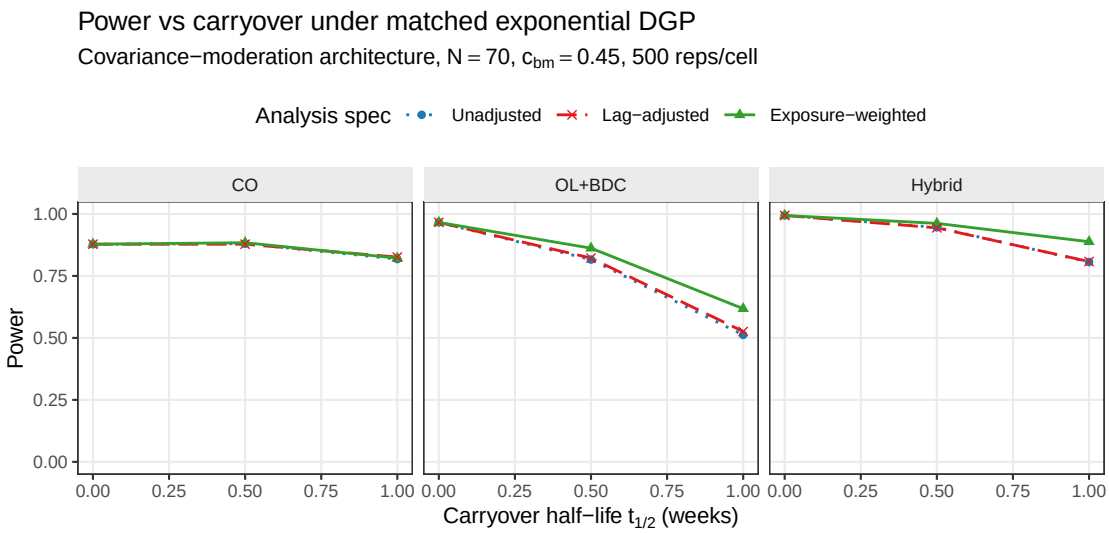


Figure 1: Stage 1: power versus carryover half-life under the exponential DGP, by analysis specification and trial design ($c_{bm} = 0.45$, $n_{sim} = 500$).

remains bm:D_{it} and the attenuation is unchanged. It marks a single occasion per path, so it cannot describe a decline across several off-drug occasions. G2 is carried no further.

Exposure weighting recovers most of the loss. At the reference cell, power under G3 falls from 0.994 with no carryover to 0.962 at $t_{1/2} = 0.5$ and 0.888 at $t_{1/2} = 1$ week. G1 gives 0.806 at $t_{1/2} = 1$. The advantage of G3 is 8 points. Its paired standard error from the per-replicate output is 0.014 (47 replicates rejected by G3 only against 6 by G1 only), so the advantage is about six paired standard errors. The advantage is zero at $t_{1/2} = 0$, where there is no carryover to model, and grows with the half-life. The relative loss of G1 from $t_{1/2} = 0$ to 1 week in this grid is 47% to 66% under OL+BDC, 19% to 56% under Hybrid and 7% to 16% under CO across N and c_{bm}. At 0.5 week the losses are 16% to 34%, 5% to 25% and -0% to 5%. The problem Stage 1 addresses is therefore a loss of up to half or more of the power under the two designs with a blinded-discontinuation phase.

The advantage depends on the design and is within noise of zero where it is not positive. At $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$, G3 leads G1 by 11 points under OL+BDC (0.618 against 0.512). OL+BDC has the fewest post-discontinuation occasions, so it has the most to gain from weighting them. G3 leads by the 8 points above under Hybrid. Under the crossover design the two are indistinguishable (0.822 against 0.818). The first off-drug occasion under CO falls 2.5 weeks after discontinuation, where less than a fifth of the effect persists, so there is little residual exposure to represent. Over the 108 alternative configurations, G3 is below G1 in 4, by at most 0.006. That difference is within the noise of a 500-replicate cell. The Weibull $k = 1$ cells repeat the exponential configurations with fresh draws. They differ from their exponential counterparts by up to 0.048 in power, which gives the scale of that noise. Earlier drafts of this manuscript reported G3 as markedly inferior to G1 under CO. That result was the pre-exposure coding defect of Section 2.8 and does not survive its correction.

Under the null the model-based test is conservative for all three specifications. The mean

Table 3: Stage 2: power of the binary indicator (G1), the exposure-weighted predictor at the true half-life (G3) and the exposure-weighted predictor at the AIC-selected half-life (G4), model-based inference, on the 30-cell corrected grid (exponential DGP, $n_{sim} = 500$, same simulated trial for all three each replicate). Two-MCSE half-width is about 0.04 at $p = 0.5$.

Design	N	c_{bm}	$t_{1/2}$	G1	G3	G4
CO	35	0.45	0.0	0.582	0.582	0.582
CO	35	0.45	0.5	0.576	0.572	0.580
CO	35	0.30	1.0	0.280	0.280	0.294
CO	35	0.45	1.0	0.514	0.524	0.532
CO	70	0.45	0.0	0.880	0.880	0.878
CO	70	0.45	0.5	0.874	0.874	0.874
CO	70	0.30	1.0	0.508	0.518	0.522
CO	70	0.45	1.0	0.832	0.852	0.844
Hybrid	35	0.45	0.0	0.826	0.826	0.816
Hybrid	35	0.45	0.5	0.656	0.678	0.688
Hybrid	35	0.30	1.0	0.212	0.250	0.276
Hybrid	35	0.45	1.0	0.450	0.524	0.534
Hybrid	70	0.45	0.0	0.992	0.992	0.990
Hybrid	70	0.45	0.5	0.960	0.960	0.962
Hybrid	70	0.30	1.0	0.448	0.518	0.534
Hybrid	70	0.45	1.0	0.790	0.886	0.876
OL+BDC	35	0.45	0.0	0.694	0.694	0.700
OL+BDC	35	0.45	0.5	0.474	0.506	0.523
OL+BDC	35	0.30	1.0	0.134	0.164	0.182
OL+BDC	35	0.45	1.0	0.248	0.330	0.344
OL+BDC	70	0.45	0.0	0.958	0.958	0.960
OL+BDC	70	0.45	0.5	0.820	0.842	0.842
OL+BDC	70	0.30	1.0	0.276	0.338	0.346
OL+BDC	70	0.45	1.0	0.544	0.638	0.639

rejection rate across the exponential null cells is 0.038 (range 0.016 to 0.058) against nominal 0.05. The conservatism is the same for each specification, so it does not disturb the comparison among them. Stage 3 addresses it. Stage 1 passes G3 forward with one caveat. Its half-life was set equal to the data-generating one, which no real analyst knows.

3.2 Stage 2: removing the analyst's half-life

Stage 2 asks whether choosing the half-life from the data by AIC (G4) recovers the power that G3 obtains at the true half-life. Table 3 reports both, with G1 as the benchmark, on the corrected 30-cell grid of Section 2.6. All three are fit to the same trial each replicate.

G4 matches G3 within Monte Carlo error at every cell. This includes the cells where G3 has the largest advantage over G1 (Hybrid and OL+BDC at $t_{1/2} = 1$). G4 does so without any pharmacokinetic input from the analyst. Under CO the three differ by at most a few points, and G4 is again indistinguishable from G3. An earlier draft found that the G4 selection drifted to the largest candidate half-life under CO and that its power collapsed. That finding was a consequence of the coding defect that affected G3 under CO and of comparing REML fits across candidates. Both are corrected (Section 2.8). Stage 2 therefore passes G4 forward

Table 4: Stage 3: conservatism of the model-based interaction test and its cluster-robust correction at four Hybrid null cells ($c_{bm} = 0$, exponential DGP, corrected simulation code, 2,000 replicates each). κ is the calibration ratio of Section 2.4; $\kappa > 1$ is a conservative test. The CR2 columns are G6, G8 and G9 for G1, G3 and G4 respectively. Two-MCSE band around 0.05: $[0.040, 0.060]$.

$t_{1/2}$	N	Specification	κ model	Type I model	κ CR2	Type I CR2
0.5	35	G1	1.05	0.041	0.97	0.049
0.5	35	G3	1.05	0.035	0.97	0.050
0.5	35	G4	1.00	0.043	0.93	0.056
0.5	70	G1	1.05	0.042	0.98	0.050
0.5	70	G3	1.06	0.037	0.99	0.050
0.5	70	G4	1.05	0.040	0.98	0.052
1.0	35	G1	1.05	0.040	0.97	0.047
1.0	35	G3	1.06	0.033	0.98	0.052
1.0	35	G4	0.99	0.046	0.91	0.059
1.0	70	G1	1.05	0.041	0.98	0.049
1.0	70	G3	1.07	0.035	0.99	0.046
1.0	70	G4	1.03	0.042	0.96	0.055

as the representation of residual exposure. Whether its selection step costs anything in size is left to Stage 3.

3.3 Stage 3: a correctly sized test

Stage 3 asks whether the interaction test is correctly sized and, if not, how to make it so without disturbing the ranking. Two findings from Stage 1 motivate it. The model-based test rejected at about 0.037 under the null. G4 adds a selection step that could in principle inflate size.

Table 4 diagnoses the conservatism at four Hybrid null cells with 2,000 replicates each. That count is enough to resolve the calibration ratio κ of Section 2.4 (two-MCSE band around 0.05: $[0.040, 0.060]$). All three Stage 1 specifications are conservative under the model-based test at every cell. G3 is the most conservative (κ 1.05 to 1.07, Type I error 0.032 to 0.037). The CR2 standard error brings κ to within 0.10 of one and Type I error inside the band for all three at all four cells. G9 is nearest the top of the band at $N = 35$ (0.056 and 0.059). Its standard error there is slightly too small (κ 0.91 to 0.93). G8 at the same cells has $\kappa = 0.98$, so the shortfall belongs to the selection step, not to CR2. At $N = 70$ G9 is at 0.052 and 0.055. E0 rejects at 0.062 to 0.064 at all four cells.

Table 7 then reports Type I error on the corrected grid for each specification with and without CR2, with E0 beside them. The grid has six null cells: three designs by two sample sizes, all at $t_{1/2} = 1$ week. It has no CR2 null cell at other half-lives outside Hybrid. Three results are visible. The model-based G1, G3 and G4 are at 0.03 to 0.05 throughout. Their CR2 variants are at 0.04 to 0.07, inside the two-MCSE band at every cell except G6 at OL+BDC, $N = 35$ (0.074, at its edge). G9 is within 0.016 of G8 at every cell and never above 0.060 in these 500-replicate cells. E0, the original analysis, rejects at 0.06 to 0.15. It is far outside the band

Table 5: Stage 3: null rejection of G6, G8 and G9 pooled over the four 2,000-replicate Hybrid null cells of Table
reftab:tab-type1-cr2; the paired difference between G9 and G8 on the same trials (the discordant counts are 77 replicates rejected by G9 only and 30 by G8 only); and G9 conditional on the half-life it selected. From '34-referee-reanalysis.R'.

Quantity	Method	n	Rate	MCSE
Pooled null rejection	G6	7986	0.0486	0.0024
Pooled null rejection	G8	7988	0.0493	0.0024
Pooled null rejection	G9	7987	0.0555	0.0026
Paired difference, G9 minus G8	G9-G8	7980	0.0059	0.0013
G9 given selected $\widehat{t}^{a}_{1/2} = 0.25$	G9	3116	0.0507	0.0039
G9 given selected $\widehat{t}^{a}_{1/2} = 0.5$	G9	2232	0.0587	0.0050
G9 given selected $\widehat{t}^{a}_{1/2} = 1$	G9	1613	0.0583	0.0058
G9 given selected $\widehat{t}^{a}_{1/2} = 2$	G9	1026	0.0585	0.0073

Table 6: Stage 3: share of replicates in which each candidate half-life (weeks) was selected by AIC, Hybrid design, pooled over every stored cell at the given true half-life and N (null and alternative cells together). From '34-referee-reanalysis.R'.

			Selected half-life			
True $t_{1/2}$	N	n	0.25	0.5	1	2
0.0	35	500	0.97	0.02	0.00	0.00
0.0	70	500	0.99	0.01	0.00	0.00
0.2	35	1000	0.95	0.05	0.00	0.00
0.2	70	992	0.98	0.02	0.00	0.00
0.5	35	34957	0.56	0.29	0.11	0.04
0.5	70	47337	0.59	0.33	0.06	0.01
1.0	35	3500	0.20	0.24	0.29	0.28
1.0	70	3493	0.12	0.26	0.40	0.21
2.0	35	1000	0.16	0.07	0.14	0.63
2.0	70	1992	0.06	0.06	0.15	0.73
4.0	35	1000	0.26	0.06	0.07	0.60
4.0	70	991	0.18	0.03	0.07	0.72

under CO and OL+BDC and at its edge under Hybrid.

The cost of the selection step. The six cells above cannot resolve a difference of one point. We therefore pooled the four 2,000-replicate Hybrid null cells of Table 4 and paired G9 with G8, the same predictor at the true half-life on the same trials. Table 5 gives the comparison. It also gives the G9 rejection rate conditional on the half-life it selected and the frequency with which each candidate was chosen.

Pooled over 7,986 fits, G9 rejects at 0.0555 (MCSE 0.0026) against 0.0493 for G8 and 0.0486 for G6. The paired difference between G9 and G8 is +0.0059 with a paired standard error of 0.0013. The selection step therefore adds about half a point of size. The G9 size under selection is about 0.055, not 0.05, and CR2 does not remove the excess. The excess is concentrated in the replicates where a half-life other than the smallest candidate was chosen (0.051 when 0.25 week is selected, 0.058 to 0.059 otherwise). Across the 48 null cells of the supplementary study outside the ladder, G9 exceeds G8 in 36, by a mean of 0.005. Table 6 shows that the

Table 7: Stage 3: Type I error at the six null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). Each model-based specification is followed by its CR2 variant (G1/G6, G3/G8, G4/G9). E0 is the original analysis. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	G1	G6	G3	G8	G4	G9	E0
CO	35	0.030	0.046	0.034	0.052	0.042	0.058	0.150
CO	70	0.046	0.054	0.046	0.050	0.050	0.056	0.128
Hybrid	35	0.042	0.066	0.030	0.044	0.042	0.060	0.070
Hybrid	70	0.032	0.046	0.034	0.054	0.038	0.058	0.060
OL+BDC	35	0.048	0.074	0.048	0.052	0.044	0.056	0.144
OL+BDC	70	0.030	0.046	0.030	0.048	0.032	0.050	0.106

selection does not recover the half-life. At a true value of 0.5 week the smallest candidate is chosen in about 56 to 59 percent of replicates and the true value in about 29 to 33 percent. At 1 week the choice is spread over all four candidates. G4 and G9 nevertheless match the power obtained at the true half-life (Table 3). The exposure weight is therefore insensitive to the half-life over this range. That insensitivity means a fixed compromise half-life with CR2 inference would avoid the selection cost altogether. The driver for that comparison is written (Reproducibility section, pending runs). Its results are not yet available.

Figures 2 and 3 show power for all nine G-family specifications on the corrected grid. The layout follows Figure 4 of Hendrickson et al. Panel A varies the biomarker effect and Panel B varies the carryover half-life.

Two features matter for what follows. First, CR2 changes power as well as size, and not in the same direction everywhere. At the Hybrid reference cell G9 reaches 0.900 against 0.876 for G4. G8 reaches 0.904 against 0.886 for G3. Over the alternative cells of the grid, adding CR2 raises power by a mean of 2.3 points under Hybrid and 3.0 under OL+BDC. Under CO it changes power by a mean of -1.1 points (minimum -5.6). The gain under the two designs with a discontinuation phase is the increase in power from correcting a conservative test. Second, G9 and G8 lead in a near-tie at every Hybrid and OL+BDC alternative cell, at both sample sizes and all three half-lives. G6 and G7 track G1 and G2 below them. Under CO all nine are within a few points of one another except G5. G9 therefore enters Stage 4 as the leading specification. It has the same power as G8 at the true half-life, size within about half a point of nominal, and no half-life required of the analyst. In what follows G6 and G8, the CR2 variants of G1 and G3, are the like-for-like power comparators for G9. G1 and G3 are retained because they are what an analyst who does not use CR2 would run.

3.4 Stage 4: external comparators on the same trials

Stage 4 asks whether the original analysis (E0) or the richer expectancy-term family (F1, F1A0B) does better than G9 on the same simulated trials. The comparison uses the scenario cells of Section 2.6. Table 8 gives Type I error at the eight null scenarios. Table 9 gives power at the five alternatives. G1, G3, G4 and G8 are retained for reference.

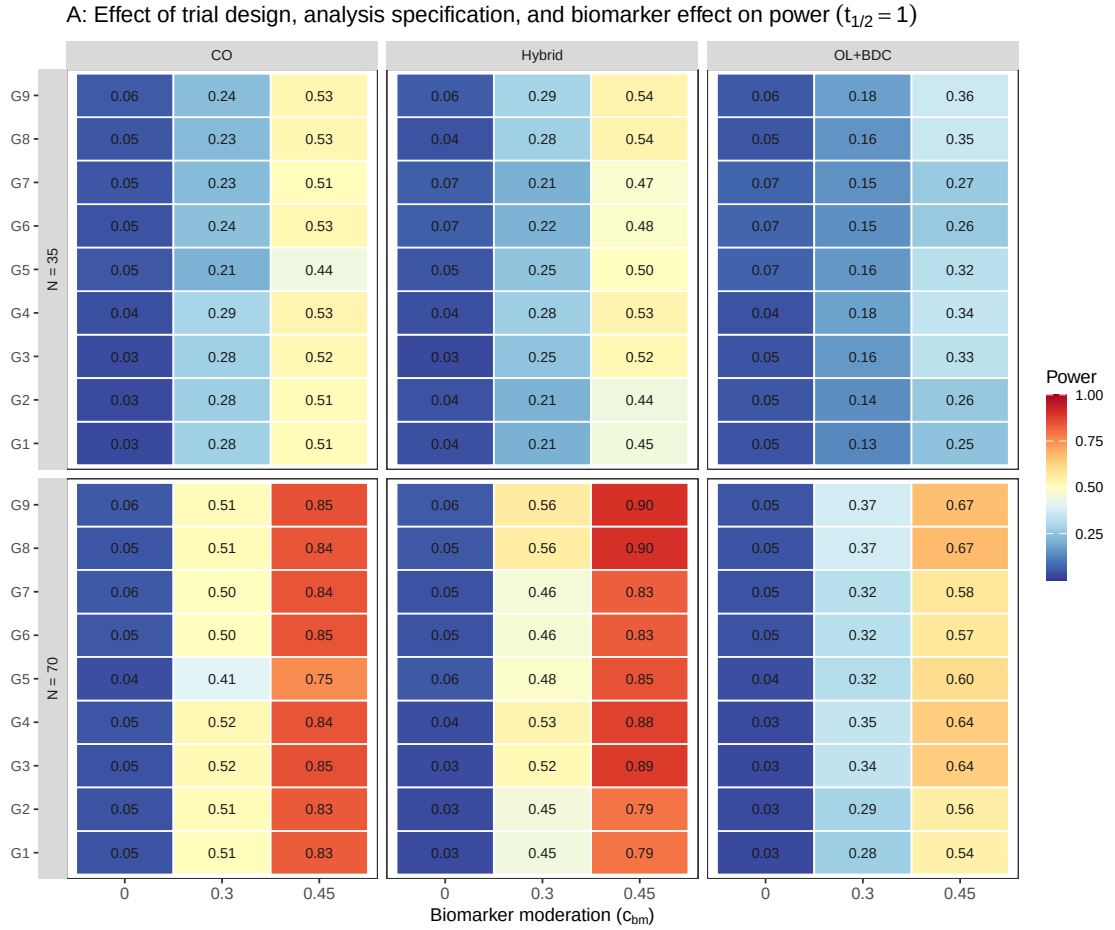


Figure 2: Stage 3: power by trial design (columns), N (row blocks) and analysis specification (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error.

Table 8: Stage 4: Type I error ($c_{bm} = 0$) at eight null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. The prognostic rows set one pathway parameter to 0.3; the prog-TV-grow row is a constant shift absorbed by the biomarker main effect and is identical to the Hybrid null row by construction. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Hybrid	0.034	0.032	0.034	0.052	0.046	0.048	0.060	0.058	0.046
OL+BDC	0.030	0.036	0.036	0.051	0.048	0.060	0.100	0.050	0.038
prog-BL 0.3	0.034	0.032	0.034	0.054	0.046	0.048	0.062	0.059	0.047
prog-TV-const 0.3	0.034	0.034	0.036	0.052	0.046	0.048	0.066	0.061	0.048
prog-TV-grow 0.3	0.034	0.032	0.034	0.052	0.046	0.048	0.060	0.058	0.046
prog-PB 0.3	0.034	0.030	0.032	0.056	0.048	0.052	0.074	0.058	0.046
$t_{1/2} = 2$ wk	0.034	0.038	0.042	0.052	0.054	0.062	0.066	0.059	0.047
Weibull $k = 2$ data	0.032	0.032	0.032	0.052	0.046	0.052	0.060	0.058	0.058

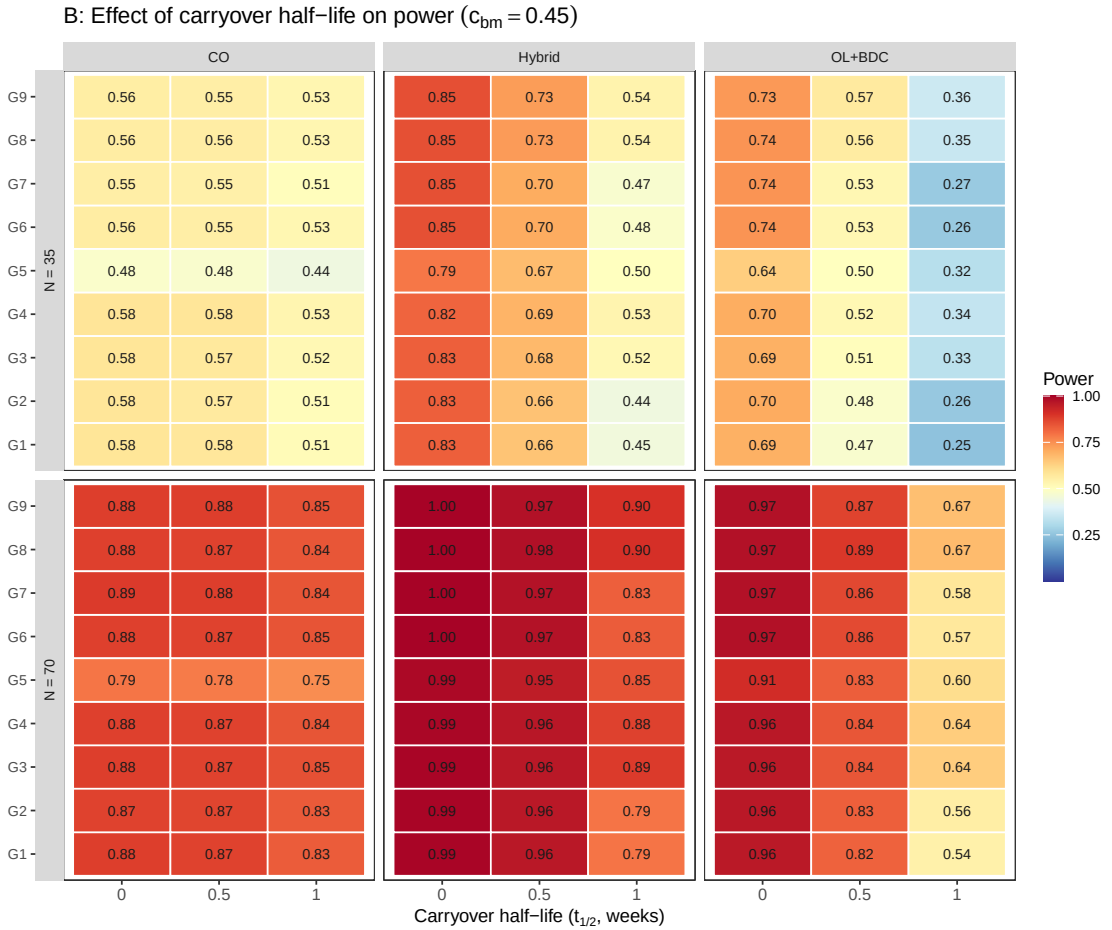


Figure 3: Stage 3: power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 9: Stage 4: power at five alternative scenarios (same specifications and n_{sim} as the null table). G6 and G8 are the equal-size comparators for G9; E0 is liberal and its power is not comparable.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Hybrid, $c_{bm} = 0.45$	0.934	0.950	0.950	0.962	0.972	0.968	0.932	0.952	0.950
Hybrid, $c_{bm} = 0.25$	0.510	0.542	0.544	0.564	0.574	0.584	0.546	0.538	0.500
OL+BDC, $c_{bm} = 0.45$	0.842	0.842	0.825	0.865	0.871	0.863	0.866	0.848	0.812
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.242	0.265	0.196	0.257	0.281	0.284	0.192	0.228
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.610	0.631	0.653	0.655	0.661	0.612	0.652	0.614

E0 is liberal under CO and OL+BDC, and at the edge of the band under Hybrid. The original analysis rejects at 0.060 to 0.100 across the eight Hybrid and OL+BDC null scenarios. It reaches 0.100 under OL+BDC. In Table 7 it reaches 0.15 under CO. In the 2,000-replicate Hybrid cells of Table 4 it is 0.062 to 0.064. Taken together, its inflation is clear under CO and OL+BDC and marginal under Hybrid, where several single cells lie inside the 500-replicate band. E0 differs from G1 in three respects (Section 2.4.6), so the comparison does not establish which of them is responsible. The residual correlation structure is the largest modeling difference and the most likely cause. The single-factor contrasts that would confirm it (E0 with `corCAR1`; G1 without) were not run. The power of E0 at the alternatives is therefore not comparable.

G9 holds size without an expectancy term at this strength. All prognostic rows leave G9 at 0.048 to 0.052, inside the band. The same holds for G8 and the F-family. The expectancy terms were introduced to protect against exactly these pathways. At this strength and sample size the G-family does not need them. Stage 5 doubles the strength and extends the sample size.

G9 loses no power to the F-family and gains where the fixed half-life of F1 is wrong. At every alternative G9 is at or above F1 (below it in 0 of 5 cells) and F1A0B (below it in 0). It is at or above its equal-size comparator G6 (below it in 1). The largest difference is at the half-life misspecification cell, where the true $t_{1/2} = 2$ weeks and F1 uses the fixed 0.2-week policy value. G9 gives 0.281 against 0.192 for F1 and 0.228 for F1A0B. The F1 half-life is below the true value at every cell of this stage (Section 2.4.6). The G9-against-F1 margin at the alternatives is therefore in part the cost of that misspecification. F1A0B selects over a grid that contains the true value, so it is the fairer comparator for the expectancy terms themselves. Under a Weibull $k = 2$ data-generating decay analyzed as exponential, G9 is again the most powerful or tied most powerful specification (0.661). The Stage 1 advantage of exposure weighting is therefore not specific to the exponential case at this effect size. The bootstrap calibration of F1A0B relative to F1 changes power by -0.038 to +0.036 across the five alternatives. It costs a few points where the fixed half-life of F1 is close to the true value and gains where it is not.

Stage 4 therefore passes G9 forward unchanged. Nothing outside the family is better sized or more powerful on the same trials.

3.4.1 Scenario grids

The supplementary study extended two of these scenarios from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. It added the within-component autocorrelation ρ as a third scenario (Section 2.6). Table 10 gives Type I error and Table 11 gives power.

The G9 Type I error lies inside the band at all fourteen null cells, from 0.034 to 0.067. It is nearest the top at the 4-week half-life, which lies above the candidate grid. These cells share random-number streams within a job and are 500-replicate cells. The result is therefore consistent with the pooled estimate of Table 5 but is not fourteen independent confirmations. The model-based G1 and G3 are conservative throughout. They are most conservative at

Table 10: Stage 4, scenario grids: Type I error ($c_{bm} = 0$) on Hybrid at $N \in \{35, 70\}$ under a Weibull decay analyzed as exponential, a true half-life below, at the boundary of and above G4's candidate grid, and two autocorrelations ($t_{1/2} = 0.5$ week and $\rho = 0.7$ unless stated; $n_{sim} = 500$; band $[0.031, 0.069]$).

Scenario	N	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Weibull $k = 0.5$ data	35	0.038	0.040	0.048	0.040	0.042	0.052	0.054	0.044	0.044
Weibull $k = 0.5$ data	70	0.026	0.028	0.028	0.034	0.040	0.048	0.060	0.042	0.050
Weibull $k = 2$ data	35	0.038	0.032	0.040	0.040	0.042	0.038	0.054	0.044	0.032
Weibull $k = 2$ data	70	0.026	0.028	0.030	0.036	0.042	0.038	0.062	0.043	0.049
$t_{1/2} = 0.2$ wk	35	0.038	0.040	0.040	0.040	0.038	0.038	0.054	0.044	0.030
$t_{1/2} = 0.2$ wk	70	0.026	0.026	0.028	0.036	0.032	0.034	0.060	0.044	0.047
$t_{1/2} = 2$ wk	35	0.036	0.022	0.034	0.042	0.050	0.056	0.048	0.046	0.028
$t_{1/2} = 2$ wk	70	0.026	0.030	0.034	0.034	0.046	0.058	0.056	0.042	0.056
$t_{1/2} = 4$ wk	35	0.036	0.022	0.044	0.042	0.044	0.064	0.050	0.046	0.030
$t_{1/2} = 4$ wk	70	0.026	0.026	0.038	0.036	0.042	0.067	0.060	0.042	0.046
$\rho = 0.5$	35	0.032	0.036	0.040	0.044	0.046	0.060	0.056	0.048	0.052
$\rho = 0.5$	70	0.024	0.030	0.028	0.030	0.044	0.046	0.062	0.044	0.034
$\rho = 0.9$	35	0.016	0.018	0.016	0.038	0.036	0.034	0.040	0.034	0.030
$\rho = 0.9$	70	0.018	0.018	0.022	0.044	0.050	0.048	0.034	0.044	0.050

Table 11: Stage 4, scenario grids: power at one alternative per scenario (same cells and n_{sim} as the null table; $c_{bm} = 0.30$ for the autocorrelation rows because 0.45 is infeasible at $\rho = 0.9$).

Scenario	N	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	35	0.538	0.552	0.572	0.548	0.586	0.588	0.586	0.541	0.523
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	70	0.878	0.894	0.889	0.900	0.915	0.913	0.876	0.891	0.891
Weibull $k = 2$ data, $c_{bm} = 0.45$	35	0.830	0.814	0.828	0.850	0.848	0.850	0.794	0.826	0.809
Weibull $k = 2$ data, $c_{bm} = 0.45$	70	0.994	0.990	0.994	0.996	0.994	0.994	0.986	0.992	0.992
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	35	0.348	0.344	0.342	0.340	0.342	0.338	0.336	0.328	0.292
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	70	0.632	0.626	0.628	0.659	0.659	0.657	0.614	0.645	0.659
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	35	0.108	0.124	0.150	0.112	0.146	0.168	0.160	0.114	0.134
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	70	0.176	0.220	0.235	0.185	0.233	0.254	0.246	0.177	0.218
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	35	0.058	0.066	0.080	0.064	0.080	0.102	0.098	0.066	0.084
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	70	0.084	0.104	0.137	0.092	0.134	0.139	0.140	0.087	0.105
$\rho = 0.5$, $c_{bm} = 0.30$	35	0.280	0.300	0.310	0.284	0.288	0.306	0.358	0.264	0.248
$\rho = 0.5$, $c_{bm} = 0.30$	70	0.542	0.546	0.552	0.595	0.606	0.596	0.610	0.522	0.456
$\rho = 0.9$, $c_{bm} = 0.30$	35	0.684	0.734	0.717	0.785	0.808	0.808	0.608	0.794	0.772
$\rho = 0.9$, $c_{bm} = 0.30$	70	0.974	0.980	0.980	0.994	0.990	0.994	0.910	0.994	0.994

Table 12: Stage 4: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$).

	Type I error				Power, $c_{bm} = 0.45$			
Dropout (rate, N)	G1	G8	G9	F1	G1	G8	G9	F1
MAR 0.1, $N = 35$	0.028	0.042	0.050	0.046	0.672	0.719	0.717	0.662
MAR 0.2, $N = 35$	0.034	0.046	0.052	0.048	0.638	0.687	0.687	0.607
MAR 0.3, $N = 35$	0.030	0.044	0.054	0.042	0.596	0.630	0.617	0.571
MCAR 0.1, $N = 35$	0.030	0.048	0.062	0.046	0.674	0.716	0.718	0.666
MCAR 0.2, $N = 35$	0.024	0.040	0.054	0.044	0.644	0.686	0.681	0.632
MCAR 0.3, $N = 35$	0.024	0.036	0.042	0.038	0.608	0.646	0.649	0.584
MAR 0.1, $N = 70$	0.026	0.040	0.040	0.042	0.956	0.980	0.976	0.958
MAR 0.2, $N = 70$	0.022	0.038	0.042	0.040	0.936	0.972	0.976	0.932
MAR 0.3, $N = 70$	0.038	0.040	0.042	0.036	0.910	0.940	0.948	0.920
MCAR 0.1, $N = 70$	0.022	0.036	0.040	0.042	0.958	0.978	0.978	0.952
MCAR 0.2, $N = 70$	0.020	0.032	0.026	0.026	0.950	0.976	0.970	0.956
MCAR 0.3, $N = 70$	0.024	0.032	0.040	0.030	0.926	0.956	0.952	0.924

$\rho = 0.9$, where the working covariance is furthest from the data. There the Type I error is 0.016 to 0.018, and κ is 1.13 to 1.18 at the null and up to 1.29 at the alternative. CR2 corrects that fully, and E0 is not liberal there. At the 18 alternative cells of the three grids, G9 is at or above F1 at every cell. It is at or above its equal-size comparator G6 at all but 4 (smallest margin -0.002). It is within 0.030 of G8 at the true half-life. It is at or above the model-based G1 at all but 3 (smallest margin -0.010). The largest margins occur where the true half-life is long (0.254 against 0.176 for G1 and 0.177 for F1 at 2 weeks, $N = 70$) and where the autocorrelation is strong (0.808 against 0.684 for G1 at $\rho = 0.9$, $N = 35$). The cells where G9 trails G1 are at the 0.2-week half-life and under Weibull $k = 2$ decay at $N = 35$. The deficit is one point or less (paired standard error 0.015 at the 0.2-week cell), within Monte Carlo error. The Weibull $k = 2$ alternatives at $c_{bm} = 0.25$ are in the summary file but not in the printed table, which shows one alternative per scenario.

3.4.2 Dropout

The scenarios so far use complete data. The last block of the supplementary study imposes monotone dropout (Section 2.6) on 10, 20 and 30 percent of patients. Dropout is either completely at random or a function of the baseline severity rank. The block runs at $N \in \{35, 70\}$ and 500 replicates (Table 12).

The G9 Type I error is 0.026 to 0.062. It is inside the band at 11 of the 12 null cells. The exception, 0.026 at MCAR rate 0.2 and $N = 70$, is below the band and therefore conservative. No cell is above the band for G9, F1 or the original analysis E0 (0.040 to 0.058). The model-based G1 is below the band at 10 of the 12 cells (0.020 to 0.038), as it was without dropout. At the alternative G9 is more powerful than G1 at all 12 cells, by 0.020 to 0.049. It is more powerful than F1 at all 12, by 0.014 to 0.081. Dropout costs power mainly at $N = 35$. There G9 falls from the 0.759 of the Stage 5 reference without dropout to 0.617 to 0.649 at rate 0.3. At $N = 70$ it falls from 0.969 to 0.948 to 0.952. The standard error of G9 is close to

Table 13: Coverage of the nominal 95 percent CR2 interval for G8 and G9 over the alternative cells of the supplementary study, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. From '34-referee-reanalysis.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
G8	35	21	0.950, 0.959	0.938
G8	70	28	0.950, 0.954	0.915
G9	35	21	0.938, 0.947	0.890
G9	70	28	0.934, 0.940	0.873

calibrated at the null (κ 0.95 to 1.05). It is somewhat too small at the alternative (κ 0.87 to 0.96). That shortfall is not a dropout effect. Across every alternative cell of the supplementary study without heavy missingness, the mean κ of G9 is 0.93 at $N = 35$ and 0.96 at $N = 70$ (minimum 0.87). G8, the same predictor and inference at the true half-life, has 1.02 and 1.01. G6 has 1.00 and 1.02. F1 has 0.99 and 1.01. The CR2 standard error without selection is calibrated under the alternative. The shortfall is specific to the data-chosen half-life and is present with complete data (Section 3.3). A nominal 95% interval for G9 will therefore cover below its nominal rate under the alternative at these sample sizes. Table 13 quantifies the amount where the per-replicate output allows it.

The G8 interval covers at its nominal rate. The G9 interval covers at about 0.93 to 0.94 on average, with a minimum near 0.87 to 0.89. The selection step therefore costs one to two points of coverage at these sample sizes.

The earlier convention of the driver, in which the rate is the per-occasion hazard, removes far more data (Section 2.6). It gives the heavy-missingness stress case. Its cells are not tabulated here. They are in `g9-gap-fill/job-31/summary.rds`. The G9 Type I error is 0.022 to 0.078. It is above the band at 1 of 12 cells (the largest, 0.078, at MCAR rate 0.3 and $N = 70$). Its κ falls to 0.49, with the worst case at MCAR rate 0.3 and $N = 35$. Power falls steeply for every specification. G9 falls to 0.083 at MCAR rate 0.3 and $N = 35$ and to 0.226 at $N = 70$. G9 exceeds G1 at 10 of 12 cells. At that level of missingness the retention of patients matters more than the choice of specification. The low κ at $N = 35$ is not a calibration failure of the kind seen under the alternative above. The estimate distribution in that cell is heavy-tailed, so the empirical SD in the denominator is dominated by a few extreme fits. The Type I error in the same cell is below nominal. A robust ratio (median standard error to a MAD-based scale) is near one. Table 14 gives the figures.

3.5 Stage 5: prognostic confounds, feasibility, and random effects

Stage 5 asks whether G9 holds size and power under a prognostic biomarker at twice the strength of Stage 4, with and without a true interaction present. It also asks whether a richer random-effects structure adds anything. It reports a finding about the simulation itself. The strength-0.6 cells on three pathways were infeasible. The robustness evidence of this stage is

Table 14: Heavy-missingness stress case (per-occasion MCAR 0.3, $N = 35$, 500 replicates): dispersion of the interaction estimates on the calibrated scale and the calibration ratio on the conventional scale (κ, mean SE over SD) and a robust one (median SE over 1.4826 times the median absolute deviation). From '34-referee-reanalysis.R'.

Cell	Method	SD of estimates	MAD-based SD	Max $\lvert\widehat{\beta}\rvert$	κ	Robust κ	Rejection
$c_{bm} = 0.45$	G6	6.1	3.1	89	0.58	0.94	0.075
$c_{bm} = 0.45$	G8	7.5	3.9	110	0.57	0.91	0.085
$c_{bm} = 0.45$	G9	12.4	5.2	137	0.46	0.78	0.083
$c_{bm} = 0.45$	F1	6.1	3.2	86	0.59	0.93	0.075
null	G6	6.1	3.3	82	0.61	0.90	0.020
null	G8	7.6	4.2	102	0.60	0.88	0.024
null	G9	12.4	5.1	131	0.49	0.86	0.022
null	F1	6.2	3.5	78	0.62	0.92	0.024

Table 15: Stage 5: Type I error of G9 and F1A0B under each prognostic pathway at confound strength 0.3, with no true interaction ($c_{bm} = 0$), Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 500$ (band $[0.031, 0.069]$). The strength-0.6 cells on prog-BL, prog-PB and prog-TV-const were generated under a repaired covariance matrix, that is, a different DGP, and are not evidence about size (see text). The constant-shift row is identical to the no-confound null by construction.

	G9		F1A0B	
Pathway	$N = 35$	$N = 70$	$N = 35$	$N = 70$
prog-BL	0.048	0.044	0.044	0.034
prog-TV-const	0.050	0.048	0.046	0.034
prog-TV-shift (constant)	0.046	0.046	0.046	0.034
prog-PB	0.046	0.044	0.046	0.034

therefore the strength-0.3 cells and the feasible ladder of Section 3.5.4. The strength-0.6 cells are reported only to document the artifact.

3.5.1 Prognostic confounds at two strengths

Table 15 gives the G9 Type I error at strength 0.3 across the three correlation pathways and the constant-shift setting at $N \in \{35, 70\}$ on Hybrid. F1A0B is shown beside it as the expectancy-term comparator. Table 16 gives the G9 power at $c_{bm} = 0.45$ at both strengths.

The G9 Type I error lies inside the band at each of the eight strength-0.3 null cells (500 replicates each, common streams within the job). Its power at strength 0.3 is within 0.039 of the no-confound reference on every pathway. That comparison is unpaired, against a cell from another job. The constant-shift setting is absorbed exactly by the biomarker main effect. It reproduces the no-confound rows to the replicate, as it must. At strength 0.6 on prog-PB and prog-TV-const, however, G9 lost 12 to 26 points of power against that reference. So did F1A0B and every other specification fitted to the same trials. The loss was uniform across fourteen specifications of three families. That uniformity was the first indication that the loss originated upstream of the analysis.

Table 16: Stage 5: power of G9 and F1A0B with a true interaction ($c_{bm} = 0.45$) and a co-occurring prognostic confound, same layout as the size table. The no-confound reference for G9 is 0.759 at $N = 35$ and 0.969 at $N = 70$ (1,000 replicates, a different job from these cells, so the comparison is unpaired). Strength-0.6 cells on prog-BL, prog-PB and prog-TV-const are a data-generating artifact, not a confound effect (see text).

	G9, $N = 35$		G9, $N = 70$		F1A0B, $N = 35$		F1A0B, $N = 70$	
Pathway	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
prog-BL	0.782	0.758	0.976	0.960	0.738	0.702	0.950	0.920
prog-TV-const	0.782	0.730	0.968	0.950	0.502	0.482	0.772	0.702
prog-TV-shift (constant)	0.778	0.756	0.970	0.966	0.778	0.756	0.970	0.966
prog-PB	0.798	0.741	0.968	0.940	0.597	0.562	0.853	0.794

Table 17: Stage 5: positive definiteness of the raw joint covariance matrix on the four Hybrid paths, and the population on-drug minus off-drug slope contrast of the outcome on the biomarker before and after the silent repair, for the prognostic cells at $c_{bm} = 0.45$ and the strength-0.6 null cells (computed from the matrix, no simulation). The contrast is the population value of the interaction term.

Pathway	Strength	c_{bm}	Paths repaired	Min. eigenvalue	Contrast, raw	Contrast, repaired
none	0.0	0.45	0	0.180	-3.26	-3.26
prog-BL	0.3	0.45	0	0.145	-3.26	-3.26
prog-BL	0.6	0.00	0	0.184	0.00	0.00
prog-BL	0.6	0.45	4	-0.110	-3.26	-3.00
prog-PB	0.3	0.45	0	0.010	-3.40	-3.40
prog-PB	0.6	0.00	4	-0.240	-0.27	-0.23
prog-PB	0.6	0.45	4	-0.642	-3.53	-2.38
prog-TV-const	0.3	0.45	0	0.011	-3.26	-3.26
prog-TV-const	0.6	0.00	4	-0.354	0.00	0.00
prog-TV-const	0.6	0.45	4	-0.853	-3.26	-1.81

3.5.2 The strength-0.6 loss is a data-generating artifact

A direct check of the covariance matrix (Table 17) shows why. Each strength-0.6 cell on prog-PB and prog-TV-const, and prog-BL at $c_{bm} = 0.45$, has a joint correlation matrix that is not positive definite on any of the four Hybrid paths. The driver then ran with repair enabled, so `corpcor::make.positive.definite` replaced the matrix by a nearby positive-definite one before any data were drawn. The repair shrinks the population on-drug minus off-drug slope of the outcome on the biomarker, which is the quantity the interaction term estimates. The slope falls from about -3.3 to -2.4 (prog-PB) and -1.8 (prog-TV-const). The fitted G9 estimates at those cells match the repaired values (-2.5 and -2.0 at $N = 35$). Their empirical standard errors fall rather than rise. The power loss is therefore the repair removing a third to a half of the interaction from the generated data. There was nothing for any analysis model to recover. Every strength-0.3 cell is positive definite, which is why none shows a loss.

Table 18: Stage 5: exact feasibility bounds (the ‘Ceiling‘ column) for each biomarker correlation (the ‘Coupling‘ column), alone and given a fixed biomarker-response correlation c_{bm}, Hybrid design (minimum over paths), $t_{1/2} = 0.5$ week, $\rho = 0.7$, $c_1 = 0.2$. Each bound is verified by the smallest eigenvalue of the full correlation matrix being positive at 0.999 times and negative at 1.001 times the stated value.

Coupling	Given	Ceiling
c_{bm}	alone	0.518
c_{pb}	alone	0.513
c_{tv}	alone	0.513
c_{bl}	alone	1.000
c_{pb}	$c_{bm} = 0.3$	0.451
c_{tv}	$c_{bm} = 0.3$	0.451
c_{bl}	$c_{bm} = 0.3$	0.815
c_{pb}	$c_{bm} = 0.45$	0.306
c_{tv}	$c_{bm} = 0.45$	0.306
c_{bl}	$c_{bm} = 0.45$	0.496

3.5.3 The feasibility bound for a prognostic correlation

The condition that failed can be computed in closed form. Write R_0 for the joint correlation matrix with the biomarker removed. R_0 depends on no biomarker correlation. Write r for the biomarker row. It is a sum of fixed patterns scaled by the correlations, $r = c_{bm}\, g + c_{pb}\, p + c_{tv}\, q + c_{bl}\, b$. By the Schur complement on the biomarker row, the joint matrix is positive definite if and only if R_0 is and

$$r^\top R_0^{-1} r = c^\top A\, c < 1, \qquad A_{jk} = x_j^\top R_0^{-1} x_k, \quad x \in \{g, p, q, b\},$$

the squared multiple correlation of the biomarker on everything else. The feasible region is the interior of an ellipsoid. For fixed values of the other correlations, the largest feasible value of any one is the positive root of a quadratic. The feasibility bound for the design is the minimum over its paths. Table 18 gives the bounds for the configuration used here. Each is verified by the smallest eigenvalue of the full matrix just below and just above it.

At $c_{bm} = 0.45$ the largest feasible prog-PB or prog-TV-const correlation is 0.306. Strength 0.3 therefore sat just inside the boundary, which corresponds to the smallest eigenvalue of 0.010 in Table 17. Strength 0.6 was infeasible at every c_{bm}. The bound for c_{bm} alone is 0.518 here. The companion study reports 0.480 for the Hybrid design. The difference arises because the bound rises with the carryover half-life (0.480 at $t_{1/2} = 0$, 0.485 at 0.2 week, 0.518 at 0.5 week under the same structure). The two agree at matched half-lives. Three consequences follow for this study and for anyone designing a similar one. First, the strength-0.6 power and size numbers must not be read as evidence about a strong confound, in either direction. The valid finding is that G9 loses no power to a prognostic confound at any feasible strength tested. Second, the driver now refuses a non-positive-definite matrix by default (Section 2.8). One exception is not a repair: a timepoint with expectancy zero has exactly zero placebo-response variance, and the positive-variance sub-block is factored instead. Third, stress scenarios should be chosen from the feasibility table before a run rather than discovered by trial and error afterwards. One

Table 19: Stage 5: rejection rate on the feasible prognostic-confound ladder (Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, $n_{sim} = 500$). Rows with $c_{bm} = 0$ are Type I error (band $[0.031, 0.069]$); the others are power. Every cell passed the pre-flight positive-definiteness check.

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
prog-PB 0.40, $c_{bm} = 0.00$	0.030	0.028	0.028	0.040	0.042	0.044	0.064	0.043	0.059
prog-PB 0.40, $c_{bm} = 0.30$	0.706	0.714	0.721	0.737	0.752	0.757	0.750	0.680	0.713
prog-PB 0.45, $c_{bm} = 0.00$	0.028	0.030	0.030	0.040	0.040	0.044	0.062	0.040	0.048
prog-PB 0.45, $c_{bm} = 0.30$	0.694	0.730	0.729	0.751	0.776	0.780	0.756	0.694	0.712
prog-TV-const 0.40, $c_{bm} = 0.00$	0.024	0.028	0.022	0.040	0.038	0.046	0.060	0.046	0.046
prog-TV-const 0.40, $c_{bm} = 0.30$	0.678	0.680	0.687	0.692	0.734	0.729	0.694	0.671	0.695
prog-TV-const 0.45, $c_{bm} = 0.00$	0.028	0.028	0.026	0.038	0.038	0.044	0.058	0.046	0.045
prog-TV-const 0.45, $c_{bm} = 0.30$	0.662	0.680	0.684	0.704	0.720	0.721	0.706	0.679	0.685
prog-BL 0.45, $c_{bm} = 0.00$	0.026	0.028	0.026	0.032	0.042	0.046	0.058	0.047	0.059
prog-BL 0.45, $c_{bm} = 0.45$	0.958	0.970	0.970	0.976	0.976	0.978	0.958	0.968	0.972
prog-BL 0.60, $c_{bm} = 0.00$	0.030	0.032	0.030	0.030	0.040	0.043	0.058	0.045	0.059
prog-BL 0.60, $c_{bm} = 0.30$	0.656	0.660	0.665	0.681	0.708	0.694	0.684	0.648	0.675
prog-BL 0.80, $c_{bm} = 0.00$	0.026	0.030	0.028	0.034	0.042	0.042	0.060	0.046	0.050
prog-BL 0.80, $c_{bm} = 0.30$	0.638	0.666	0.664	0.679	0.688	0.682	0.692	0.650	0.660

qualification applies to the whole stage. The bound applies to prognostic effects expressed as correlations of a single latent biomarker with the components. It is a property of this parameterization. A prognostic effect expressed as a mean shift, such as a biomarker that predicts the rate of natural-history change, has no such bound. It can be as strong as the clinician believes. The finding that nothing changes inside the feasible region is a statement about the correlation pathways, not about confounding in general.

3.5.4 Robustness up to the feasibility bound

The supplementary study then ran a ladder of feasible confounds chosen from Table 18. The ladder sets c_{pb} and c_{tv} at 0.40 and 0.45 (bound 0.451 at $c_{bm} = 0.30$), c_{bl} at 0.6 and 0.8 (bound 0.815), and $c_{bl} = 0.45$ at $c_{bm} = 0.45$ (bound 0.496). Each is run with and without a true interaction, at $N = 70$ and 500 replicates (Table 19).

The G9 Type I error is 0.042 to 0.046 across the seven null cells. Its power at $c_{bm} = 0.30$ is 0.682 to 0.780. It does not fall with strength on any pathway. prog-PB gives 0.757 at 0.40 and 0.780 at 0.45, prog-TV-const 0.729 and 0.721, prog-BL 0.694 and 0.682. The mean interaction estimate is the more exact check, because the repair artifact of Table 17 appeared as attenuation. The G9 mean estimate at $c_{bm} = 0.30$ is -2.37 to -2.35 under prog-TV-const and prog-BL at both strengths of each, and -2.45 to -2.43 under prog-PB. The prog-PB value is the same estimate plus the small expectancy-scaled contrast discussed below. No strength moves it. At $c_{bm} = 0.45$ with $c_{bl} = 0.45$, the G9 power is 0.978 against 0.969 with no confound (the reference of Table 16). Its mean estimate is -3.53. The loss that Stage 5 first attributed to a strong confound does not appear anywhere inside the feasible region on the grid run. The strongest tested correlation lies 0.001 below the bound for prog-PB and prog-TV-const at $c_{bm} = 0.30$ (0.45 against 0.451). It lies 0.015 below for prog-BL at $c_{bm} = 0.30$ (0.8 against 0.815). It lies 0.046 below for prog-BL at $c_{bm} = 0.45$ (0.45 against 0.496). prog-PB and prog-

Table 20: Stage 5: Type I error under prog-PB with no true interaction at larger samples (Hybrid, $t_{1/2} = 0.5$ week; $n_{sim} = 1{,}000$ at $N = 140$ and 280, band $[0.036, 0.064]$; the $N = 70$ row is from the ladder, $n_{sim} = 500$).

Cell	G1	G3	G4	G6	G8	G9	E0	F1	F1A0B
$c_{pb} = 0.30$, $N = 140$	0.044	0.043	0.043	0.058	0.055	0.055	0.068	0.057	0.049
$c_{pb} = 0.30$, $N = 280$	0.053	0.045	0.052	0.065	0.063	0.063	0.071	0.067	0.060
$c_{pb} = 0.45$, $N = 70$	0.028	0.030	0.030	0.040	0.040	0.044	0.062	0.040	0.048
$c_{pb} = 0.45$, $N = 140$	0.046	0.042	0.045	0.059	0.056	0.059	0.072	0.057	0.046
$c_{pb} = 0.45$, $N = 280$	0.055	0.049	0.054	0.066	0.062	0.062	0.077	0.067	0.061

TV-const were not run above 0.3 at $c_{bm} = 0.45$, where the bound is 0.306. The no-confound references are from other jobs, so these comparisons are unpaired.

The last question is whether the prog-PB contrast, which is too small to detect at $N \leq 70$, becomes Type I error inflation as N grows. Table 20 gives the null rejection rate at $N \in \{140, 280\}$ and $c_{pb} \in \{0.3, 0.45\}$ at 1,000 replicates (band $[0.036, 0.064]$). The F-family, which carries the expectancy terms, is the comparator.

At $N = 280$ and $c_{pb} = 0.45$, G9 rejects at 0.062 and F1 at 0.067. At $c_{pb} = 0.3$ the figures are 0.063 and 0.067. The model-based G1 is at 0.053 to 0.055. At $N = 280$ the CR2 tests and F1 sit at 0.062 to 0.067, at or just above the 1,000-replicate band. The rejection rate of G9 rises with N (0.044 at 70, 0.059 at 140, 0.062 at 280 for $c_{pb} = 0.45$). The expectancy terms do not separate F1 from G9. The F1 mean estimate is nearer zero (0.04 against -0.04 for G9 at $c_{pb} = 0.45$), and both standard errors are calibrated at $\kappa = 0.96$. The study has no confound-free cell at $N = 280$, so the excess is not attributed. A small bias in the estimand of a fifth to a third of a standard error would produce about 0.06. A rise with N is what such a bias would produce. The population on/off contrast of the next section (-0.14 at $c_{pb} = 0.3$) is on the raw outcome scale. The G9 coefficient on D_{bc} is not. The two are therefore not directly comparable and have not been reconciled. The expectancy-term family offers no protection that G9 lacks at any N examined. Neither family is shown to hold size at $N = 280$ under this pathway.

3.5.5 Random-effects augmentation adds nothing under this DGP

Before the artifact was identified, the strength-0.6 loss had been read as added patient-level variance that a random intercept could not absorb. G9 was therefore refitted with a random slope on time, on expectancy, and on both, on twelve of the Stage 5 cells at 1,000 replicates. Eight of those cells are the repaired strength-0.6 cells. Table 21 shows the result. All three variants match plain G9 to within one point of power and of standard error at every cell. The paired per-replicate differences in rejection are at most 0.008, from the per-replicate output. The three-component structure has up to 2.5% non-convergence. This is the expected result once the mechanism is understood. It is also what the data-generating process implies. The DGP contains no patient-specific slope, so there is no variance for a random slope to absorb. The check therefore shows that adding random slopes costs nothing here. It does not show that random slopes would add nothing in a trial whose patients differ in their trends. That

Table 21: Stage 5: rejection rate of G9 and its three random-effects variants on twelve cells (Hybrid, $t_{1/2} = 0.5$ week, $n_{sim} = 1{,}000$; $c_{bm} = 0$ rows are Type I error). The strength-0.6 rows were generated under the repaired matrix described above and are not evidence about the confound.

Cell	G9	+ slope on t	+ slope on D_e	+ both
none, $c_{bm} = 0.00$, $N = 35$	0.050	0.050	0.050	0.046
none, $c_{bm} = 0.00$, $N = 70$	0.043	0.044	0.043	0.041
none, $c_{bm} = 0.45$, $N = 35$	0.759	0.762	0.766	0.769
none, $c_{bm} = 0.45$, $N = 70$	0.969	0.970	0.969	0.970
prog-PB 0.6, $c_{bm} = 0.00$, $N = 35$	0.060	0.060	0.059	0.060
prog-PB 0.6, $c_{bm} = 0.00$, $N = 70$	0.047	0.046	0.047	0.047
prog-PB 0.6, $c_{bm} = 0.45$, $N = 35$	0.594	0.593	0.593	0.596
prog-PB 0.6, $c_{bm} = 0.45$, $N = 70$	0.860	0.859	0.858	0.858
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 35$	0.050	0.053	0.051	0.054
prog-TV-const 0.6, $c_{bm} = 0.00$, $N = 70$	0.044	0.044	0.043	0.040
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 35$	0.489	0.488	0.489	0.485
prog-TV-const 0.6, $c_{bm} = 0.45$, $N = 70$	0.772	0.772	0.773	0.768

case needs a DGP with a patient-level slope variance, which was not simulated.

4 Discussion

4.1 What each stage established

The sequence of stages was not planned in advance. Each stage was built to answer the question the previous one raised. The recommendation rests on the chain of stages rather than on any single comparison. Everything below is conditional on the correlation encoding of the companion study, on one parameter set calibrated to prazosin and PTSD trajectories, and on the Hybrid design. The analysis model is close to the conditional mean of that Gaussian DGP. Part of what the stages show is therefore that a correctly specified regressor outperforms a misspecified one. The findings concern which choices cost power or size under that DGP, not the DGP itself.

Stage 1 established that the choice of exposure representation is a question of power, not validity. Only the representation that changes the regressor carrying the interaction recovers any power. The lagged-treatment term, the classical remedy, changed nothing measurable across 162 configurations. The reason is structural. It enters as a main effect and leaves the attenuated interaction unchanged. A practitioner should not report a lagged term as an adjustment for carryover. Exposure weighting recovered most of the loss under the two designs with a blinded-discontinuation phase. It was within Monte Carlo noise of the binary indicator where it did not gain. The earlier finding that it failed under a crossover design was a coding defect, and we withdraw it. Under this DGP the exposure-weighted predictor at the true decay is the correctly specified regressor. Stage 1 therefore measures the cost of the two misspecifications, not a general superiority.

Stage 2 established that the true half-life, which Stage 1 gave the analyst, is not needed for power. AIC selection over four candidates matches the true half-life within Monte Carlo error at every cell of the corrected grid. This includes the crossover design, once the selection compares ML fits and the predictor is coded correctly before first exposure. Stage 2 does not establish that the selection recovers the half-life. The selected value is spread over the candidates (Table 6). The power match means the exposure weight is insensitive to the half-life over this range. That insensitivity suggests a fixed compromise value with CR2 inference as an alternative that avoids selection altogether. It has not been evaluated here.

Stage 3 established three results. The model-based test is conservative by about a quarter of its nominal size. The conservatism comes from a working covariance that does not match the data. A CR2 cluster-robust standard error with Satterthwaite degrees of freedom brings every specification to within about half a point of nominal size. It raises power by two to three points on average under the Hybrid and OL+BDC designs and lowers it slightly under the crossover design. Stage 3 also measured what the selection step costs. Paired against the same predictor at the true half-life, the G9 size is about half a point above the G8 size. The G9 standard error under the alternative is a few percent too small. This is an effect of selection and not of CR2 (Section 3.4). The cost is small and is stated as a cost.

Stage 4 established two results about analyses outside the family. The original analysis is liberal under the crossover and OL+BDC designs, by a factor of two to three, and marginally liberal under Hybrid. It differs from the G-family in three respects. The missing residual correlation structure is the largest of them. The single-factor contrast that would attribute the inflation was not run. The expectancy-term family was built to protect against a prognostic biomarker. It provides no size that G9 lacks at the strengths and sample sizes examined. Its fixed-half-life member loses power where the half-life is misspecified. At the half-lives of this study, that is everywhere. Its bootstrap-calibrated member gains or loses a few points depending on the cell. The scenario grids confirmed the ordering across sample size and effect size. G9 was below G1 by one point or less at two $N = 35$ cells, within Monte Carlo error. Dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, changed neither the size of G9 nor its ordering against G1 and F1. Only the heavier per-occasion convention, which removes up to 73 percent of the observations, collapsed power.

Stage 5 established that G9 holds size under the three correlation pathways at strength 0.3 and on the feasible ladder. It loses no power and no part of its estimate to a confound inside the feasible region on the grid run. It gains nothing from random slopes under a DGP that has none. The strength-0.6 cells are evidence about a repaired DGP and not about the confound. A biomarker correlated with the placebo response does not inflate the G9 Type I error beyond that of the expectancy-term family at N up to 280. At $N = 280$, however, both families sit at or just above the band, and the excess is not attributed. The pathway most likely to bias the estimator, a biomarker that predicts the rate of natural-history change, was not simulated. The stage makes two methodological contributions. It identifies a repaired non-positive-definite covariance matrix as the source of an apparent robustness failure. It gives the closed-form feasibility bound that allows a stress grid to be designed inside the valid region.

4.2 The recommended analysis for a Hybrid trial

For a Hybrid aggregated N-of-1 trial under the correlation encoding we recommend G9:

$$\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}(\hat{t}^a_{1/2}) + bm_i{:}D_{bc,it}(\hat{t}^a_{1/2}),$$

with a patient random intercept and `corCAR1` residual correlation. $D_{bc,it}$ equals one on drug, $2^{-t_{sd}/\hat{t}^a_{1/2}}$ off drug after first exposure, and zero before first exposure. $\hat{t}^a_{1/2}$ is chosen by ML-AIC over $\{0.25, 0.5, 1, 2\}$ weeks with the interaction in the selection model, and the selected model is refitted by REML. The interaction is tested with a CR2 cluster-robust standard error and Satterthwaite degrees of freedom. Three properties recommend it over the alternatives considered. It requires no pharmacokinetic input from the analyst and matches the power of the predictor at the true half-life. Its size is within about half a point of nominal at the sample sizes examined. The excess is the measured cost of the selection step. It holds that size under the three prognostic correlation pathways at every feasible strength tested, without an expectancy term. It is at least as powerful as the richer expectancy-term family at every alternative scenario, with a clear advantage when the true half-life is long. Two qualifications apply. Its nominal confidence interval is slightly narrow under the alternative, by the few percent that selection costs. An analyst who needs an interval rather than a test should prefer an alternative that avoids selection. Two such alternatives are a fixed compromise half-life with CR2, and selection on the model without the interaction followed by a bootstrap calibration as in F1A0B. Its size has been measured only up to $N = 70$ with complete data and at $N = 280$ under one pathway, where it sits at the edge of the band.

The recommendation transfers to OL+BDC on the evidence of Stages 1 to 4, where G9 and G8 lead in the same near-tie. Under the crossover design the specifications are within a few points of one another, and G9 is not worse than any. The choice there is immaterial on this evidence, and the expectancy-term family is not identifiable at all. The recommendation is stated for the correlation encoding.

4.3 What the expectancy terms are for

The G-family omits the design-expectancy covariate that the F-family carries. Stages 4 and 5 show that G9 does not need it for size under any pathway at the strengths and sample sizes examined. One mechanism deserves a note. The placebo-response amplitude scales with expectancy. Expectancy differs between on-drug and off-drug visits in the Hybrid design. A biomarker correlated with the placebo response therefore acquires a small on/off slope difference by construction. On the raw on/off scale of Table 17, the population contrast under prog-PB is -0.14 at strength 0.3 and -0.27 at 0.6 with no true interaction. The contrast at the alternative is -3.3. The G9 coefficient on D_{bc} is on a different scale. Its mean estimates at $N = 280$ (-0.013 at $c_{pb} = 0.3$, -0.039 at 0.45) have not been reconciled with the population contrast. This contrast is exactly what D_e and $bm{:}D_e$ absorb. At N up to 280 and c_{pb} up to 0.45 it produced no inflation in G9 beyond that of F1, which carries the terms (Section 3.5). F1 matched the G9 power at every prog-PB alternative. Both sat at the edge

of the band at $N = 280$. A targeted size check at larger N with a confound-free counterpart should resolve that. Until then, the robustness of either family to a biomarker correlated with the placebo response should not be stated without qualification. A trial with a biomarker known to track placebo responsiveness loses nothing by adding the two terms. The evidence here neither requires nor excludes them.

4.4 Limitations

The numbers come from a single parameter set calibrated to a prazosin and PTSD trajectory. That set has one trajectory shape per component, one response SD vector, $\rho = 0.7$ except in the ρ grid, and $c_1 = 0.2$. The recommendation comes from one design, and the power results from $N \leq 70$. Absolute power values are therefore illustrative. The ranking, conditional on design, is the more portable result. The correlation encoding is itself an unvalidated modeling assumption of this project. The analysis model is close to its conditional mean. The study therefore cannot show how the ranking would change under a DGP with a different mean structure. The selection step costs about half a point of size and a few percent of standard-error calibration under the alternative. Coverage of the CR2 interval has been quantified only where the per-replicate output allows (Section 3.4). A selection-aware interval has not been evaluated. Dropout was examined at 10 to 30 percent of patients, monotone, with the MAR mechanism tied to baseline severity. Baseline severity is not among the analysis variables, so the mechanism is MAR given the analyzed data only insofar as early outcomes proxy for it. Informative dropout related to the outcome or to treatment response was not simulated. The dropout block has no alternative cell without dropout under the same code. Under the heavier per-occasion convention power collapsed for every specification. The recommendation is therefore for trials that retain most of their observations. The scenario grids use one alternative effect size per scenario, and the prognostic ladder uses one sample size. The prognostic pathways are correlations of a single latent biomarker with the components. The feasibility bound is a property of that parameterization. A biomarker that predicts the rate of natural-history change is a mean shift with no such bound. It was not simulated, and it is the pathway most likely to bias an estimator without a biomarker-by-time term. The feasibility bounds are specific to the Hybrid design at $t_{1/2} = 0.5$ week, $\rho = 0.7$ and $c_1 = 0.2$. They must be recomputed for any other configuration, which the closed form makes immediate. Size and power at the strength-0.6 prognostic cells were measured under a repaired matrix. They are results about a modified data-generating process. The large-sample prog-PB cells have no confound-free counterpart at the same N. The excess over nominal seen there for the CR2 and F-family tests at $N = 280$ is therefore not attributed. Cells within a job share random-number streams. The no-confound references for Stage 5 come from other jobs. Counts of cells inside a band are therefore not independent, and the cross-job comparisons are unpaired. The calibration ratio κ uses the mean rather than the root mean square of the standard errors. It is biased downward by a few percent at $N = 35$. The random-effects check was run under a DGP without patient-specific slopes. The effect-size curve of earlier drafts was not rerun under the corrected code. The sensitivity checks vary one factor at a time, so they can miss interactions between factors.

4.5 Related approaches not evaluated

Three N-of-1 analysis approaches sit outside the family examined. Each would require substantial independent development to enter the comparison. Tang and Landes (20) propose formula-based t-tests that correct a single-subject paired-difference test for an estimated serial correlation. They have no biomarker-moderation term. Schmid and Yang (21) describe a Bayesian hierarchical alternative with partial pooling of individual-level treatment, trend and autocorrelation parameters. They note that carryover is tractable only in aggregated analyses like ours. Daza (22) frames N-of-1 inference as a counterfactual time-varying-treatment problem. In that framing carryover is confounding to be controlled rather than a predictor to be modeled. Chen and Chen (3) found that a paired t-test matched or exceeded mixed models in simulated N-of-1 series with and without carryover. Araujo, Julious and Senn (23) describe the variance components that a set of N-of-1 trials identifies. These two papers motivated the paired-difference specification G5 in earlier drafts. On the corrected grid G5 trails the mixed-model family under every design and is not pursued. The closest prior work on an exponentially decaying exposure term is the carryover-adjusted parametric model of Gärtner et al. for aggregated N-of-1 trials, with exponential wash-in and wash-out (6). Its estimand is the treatment effect, not the biomarker interaction. The biomarker interaction is where this study's contribution lies. Blackston et al. (5) show inflated Type I error for aggregated N-of-1 analyses that ignore moderate or strong carryover. That result is consistent with the E0 and G1 results here. Percha et al. (4) give design recommendations from simulated single-series N-of-1 trials with carryover and wash-in. Zucker et al. (24) set out the methodology for combining N-of-1 trials into population estimates. The case against a single lagged term as a carryover adjustment in crossover trials is classical (25,26).

5 Conclusions

1. **Represent residual exposure in the regressor, not as a nuisance term.** The classical lagged-treatment term never improved on ignoring carryover, on any measure, in any design. An exposure-weighted predictor recovers most of the power that carryover removes under the Hybrid and OL+BDC designs. It is within Monte Carlo noise of the binary indicator where it does not gain.
2. **Let the data choose the half-life, or fix a compromise value.** AIC selection over four candidates, comparing ML fits and refitting the selected model by REML, matches the power of the predictor at the true half-life at every cell. It removes the need for the analyst to supply a half-life. The selected value is spread over the candidates. The match therefore reflects the insensitivity of the predictor to the half-life rather than recovery of it.
3. **Use a CR2 cluster-robust standard error, and state the cost of selection.** The model-based test rejects at about 0.037 under the null. CR2 with Satterthwaite degrees of freedom brings every specification to within about half a point of nominal size. It raises power by two to three points under the Hybrid and OL+BDC designs, but not under the crossover design. Paired against the predictor at the true half-life, selection

adds about half a point of size and makes the standard error a few percent too small under the alternative.

4. **Do not use the original analysis as is.** It is liberal by two to three times nominal under the crossover and OL+BDC designs and marginally liberal under Hybrid. It differs from the G-family in its residual correlation structure, an expectancy term and its software. The first is the likely cause.
5. **Expectancy terms are not needed for size at the strengths and sample sizes examined, and their fixed-half-life member costs power.** G9 holds nominal size under the three prognostic correlation pathways at every feasible strength examined, up to within 0.001 to 0.046 of the feasibility bound. It is at least as powerful as the bootstrap-calibrated expectancy-term specification at every alternative. At $N = 280$ under a biomarker correlated with the placebo response, both families sit at the edge of the band. A biomarker that predicts the rate of natural-history change was not simulated.
6. **Check feasibility before stressing the data-generating process.** A prognostic correlation that exceeds the closed-form feasibility bound yields a non-positive-definite matrix. Repairing that matrix without warning removes part of the interaction and produces a spurious robustness failure in every analysis model at once. The bound is a quadratic root and should be computed first. It bounds correlations only. Mean-shift confounds are unbounded.
7. **Recommendation.** For a Hybrid aggregated N-of-1 trial under the correlation encoding, use G9: the exposure-weighted predictor at the AIC-selected half-life with CR2 inference. Under dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, it keeps its size and its power advantage. It should be accompanied by an effort to retain patients, because heavier missingness costs every analysis power.

Glossary

- **Analysis codes (G1 to G9, E0, F1, F1A, F1A0, F1A0B).** The analysis specifications of Table 1 and Section 2.4.6. Stored data files label the G-family by `E`-prefixed codes (`E1`, `E3`, `E2`, `E7`, `E9` and their `cr2` variants for G1 to G9 respectively), displayed as G codes at render time.
- **Biomarker-response correlation (c_{bm}).** The on-drug correlation between the biomarker and the drug-response component. It is the interaction estimand of this study. Stored data and code refer to it as `c_bm` or `coupling`.
- **Carryover half-life ($t_{1/2}$).** The time after discontinuation at which the residual drug effect, and under the correlation encoding the biomarker-response correlation, has fallen to half its on-drug value. The analysis half-life $t^a_{1/2}$ is the value the analyst assumes or selects.
- **Half-life policy value.** The fixed analysis half-life of 0.2 week used by F1, set by project policy.
- **Common random numbers.** Fitting every specification to the same simulated dataset

within a cell, so that differences between specifications are paired and have reduced Monte Carlo variance.

- **CO (crossover).** The two-period blinded crossover design of Section 2.2.
- **CR2.** The bias-reduced cluster-robust sandwich standard error of Bell and McCaffrey, which estimates the variance of the interaction coefficient from between-patient variation.
- **Correlation encoding of the interaction.** The data-generating convention in which the biomarker-treatment interaction is a correlation between the biomarker and the drug response that is present on drug and absent or decayed off drug. Code and data files label it `mvn` or covariance moderation.
- D_{bc}**.** The exposure-weighted predictor: one on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, zero before first exposure.
- **Expectancy (**D_e**,** η_t**).** The participant's probability of being on drug by design: one during open-label periods and one half during blinded periods. It scales the placebo-response component.
- **Extended DGP.** The data-generating process of the companion study, in which the biomarker-response correlation decays after discontinuation, the within-component correlation is separable AR(1), carryover decays from the last on-drug mean, and a non-positive-definite covariance matrix stops the simulation. Code and data labels call it the refined process.
- **Feasibility bound.** The largest value of a biomarker correlation for which the joint correlation matrix is positive definite on every path of the design, computed from the Schur complement (Section 3.5.3). Table 18 labels it Ceiling.
- **Hybrid design.** Open-label titration, blinded discontinuation, then a brief crossover (Section 2.2).
- κ **(calibration ratio).** The mean model standard error divided by the empirical standard deviation of the estimates across replicates. A value of one is correctly calibrated.
- **MCSE.** Monte Carlo standard error, the standard error of a simulation estimate that arises from the finite number of replicates.
- **Natural-history component (TV).** The latent component of the outcome that follows the untreated time course.
- **OL+BDC.** Open-label titration followed by blinded discontinuation (Section 2.2).
- **Original DGP.** The data-generating process published by Hendrickson et al. (2020), from which the extended DGP is derived.
- **Original analysis (E0).** The analysis published with the original DGP: `lme4::lmer` with a random intercept, no residual correlation structure, a binary on-drug indicator, and Satterthwaite p-values.
- **Paired-difference specification (G5).** A specification that collapses each patient's series to one on-minus-off difference and regresses it on the biomarker by ordinary least squares.
- **Placebo-response component (PB).** The latent component of the outcome that responds to expectancy rather than to the drug.
- **Prognostic correlation pathways (prog-BL, prog-TV-const, prog-PB, prog-**

TV-grow). Correlations of the biomarker with baseline severity (c_{bl}), with the natural-history component (c_{tv}), or with the placebo-response component (c_{pb}), and a constant mean shift of the natural-history component (b_{tv}), respectively (Section 2.4).

- **Satterthwaite degrees of freedom.** An approximate degrees of freedom for the t reference distribution of a cluster-robust test, computed from the variance of the variance estimate.
- **Separable AR(1) structure.** The same first-order autoregressive correlation for every component across occasions, with a constant between-component correlation at every lag.
- **Supplementary study.** The simulation study specified after Stage 5 that closed the items the stages left open. Code and data files label it `gap-fill`.
- **Two-MCSE band.** The interval of plus or minus two Monte Carlo standard errors around the nominal 0.05 rejection rate, which is $[0.031, 0.069]$ at 500 replicates, $[0.036, 0.064]$ at 1,000 and $[0.040, 0.060]$ at 2,000.

6 Declarations

Ethics approval and consent to participate. Not applicable. This is a simulation study with no human participants or patient data.

Consent for publication. Not applicable.

Availability of data and materials. The data and code supporting the findings of this study are available in the pmsimstats-ng project repository (`https://github.com/rgt47/pmsimstats-ng`). A tagged release with a persistent identifier will be deposited at submission. The commit reported here should be cited until then. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths are listed in the Reproducibility section below. The stored summaries also contain cells from a mean-shift encoding of the interaction (`meanmod` and `B3-meanmod` labels) and a linear decay form. Both are outside the scope of this manuscript. The data dictionary in the repository marks them as such.

Competing interests. To be completed by the authors. Any author of this paper who is also an author of the published framework (1) must be disclosed here.

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Reporting. This is a simulation study and reports no trial. The CENT 2015 extension of CONSORT (27) governs the reporting of any N-of-1 trial that applies the recommended analysis.

7 Reproducibility

This manuscript was rendered inside the project's zzcollab Docker container, with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulations ran under R 4.6.0 on the project's simulation host. The study uses `nlme` for the continuous-time linear-mixed analysis, `clubSandwich` for CR2 inference, `lme4` and `lmerTest` for E0, and `parallel` for the parameter-grid sweep. No simulation code runs during knitting. Knitting loads pre-computed summaries only.

Drivers and data, by stage. All paths are under `analysis/scripts/carryover-sensitivity/` (scripts) and `analysis/data/` (summaries).

- Stage 1: `01-run-factorial.R` (540 cells, 500 replicates, master seed 20260415; rerun 2026-10-25 under the corrected code), `02-summarise-grid.R` producing `02-grid-summary.rds`, and `03-render-figures-extra-slim.R` producing Figure 1.
- Stages 2 and 3: `27-run-tier1-g-vs-f-e0.R` (30 cells, 500 replicates, 2026-10-25) producing `carryover-sensitivity/tier1-g-vs-f-e0/summary.rds`, from which Tables 3 and 7 and Figures 2 and 3 (via `12-hendrickson-heatmap.R`) are drawn. The size diagnostic of Table 4 is block A of the supplementary study below.
- Stage 4: `06-g-vs-f-comparison.R` (13 of its 15 cells are reported; 500 replicates, 2026-10-23) producing `carryover-sensitivity/g-vs-f-comparison/summary.rds`.
- Stage 5: `28-run-g9-prognostic-stress.R` (32 cells, 500 replicates, 2026-10-25) and `29-run-g9-random-effects.R` (12 cells, 1,000 replicates, 2026-10-26) producing `carryover-sensitivity/g9-prognostic-stress/summary.rds` and `carryover-sensitivity/g9-random-effects/summary.rds`; `30-check-prognostic-confound-pd.R` and `31-prognostic-coupling-ceiling.R` producing `pd-check.rds` and `coupling-ceilings.rds` in the former directory, each running in seconds with no simulation.
- Supplementary study (labeled `gap-fill` in code and data): `32-run-g9-gap-fill.R` under the working protocol `32-gap-fill-protocol.md`, run as four simrun jobs (blocks A, B and C; jobs 31, 32 and 33 on the primary host and job 40 on a second, 2026-10-26 to 2026-10-27, at commit `ca2eef2` or a code-identical successor, except job 33, which reran the dropout cells at commit `fc7ee08` under the patient-level convention), assembled by `33-assemble-gap-fill.R` into `carryover-sensitivity/g9-gap-fill/summary.rds` with each job's own outputs under `g9-gap-fill/job-NN/`. Tables 4, 10, 11, 12, 19 and 20 are drawn from it. The per-occasion dropout cells of job 31 are in the same file. The pre-flight eigenvalue of every cell is in `preflight.rds`.
- Corrections: the paired check of the pre-exposure coding defect is `26-co-dbc-fix-paired.R` (`carryover-sensitivity/26-co-dbc-fix-paired-reps500.rds`); the cross-factor constant correction and the stop-on-non-PD default are in `simulation-core.R` and `implementations/tidyverse/R/functions.R`.
- Reanalysis of the per-replicate output for the 2026-10-27 referee report (pooled and paired size of G9 and G8, selection frequencies, calibration and coverage at the alternatives, the heavy-missingness dispersion): `34-referee-reanalysis.R` producing the CSV files in `carryover-sensitivity/referee-reanalysis/`, from which Tables 5, 6,

13 and 14 are drawn.

- Pending runs: `36-run-referee-variants.R` (checked by `35-check-new-variants.R`) adds, with independent seeds per cell,

  (i) a null-model selection variant of G9 (half-life chosen without the interaction, stored as `E7null_cr2`), (ii) G8 at fixed wrong half-lives of 0.25 and 1 week (`E3fix_cr2`), (iii) a biomarker-by- trend prognostic pathway, a natural-history mean shift growing with the component's time course (`b.bm.tv.trend`, unbounded by the correlation matrix), and (iv) a patient-specific slope variance (`sd_slope`), at 5,000 replicates for the null cells that carry the recommendation and 1,000 for the alternatives. The local estimate for the simulation host is about 8 to 10 hours. Its results are not in this version of the manuscript. The claims they bear on (the cost of selection against a fixed compromise half-life, size under a trend pathway, and the random-slope check on a DGP with slopes) are stated above as unresolved.

The fitting functions for every specification (`fit_spec_s7`, `cr2_extract`, `select_half_life_aic`, `fit_spec_e0`, `fit_f_family`, `simulate_cell_combined`, `simulate_cell_g9_random`) are in `simulation-core.R`. Specifications are stored under `E`-prefixed codes (`E1`, `E2`, `E3`, `E7`, `E9` and their `cr2` variants) in every data file and displayed as `G1-G9` at render time by `spec-labels.R`. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`. The migration is documented in `analysis/report/NOTATION.md`. The exploratory sensitivity blocks of earlier drafts and their figures are documented in `supplement.Rmd` and the `archive/` directory of this manuscript. The scripts that produced them are retained under the same directory as above.

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